ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Structure Building and Coding

I. Materials Studio

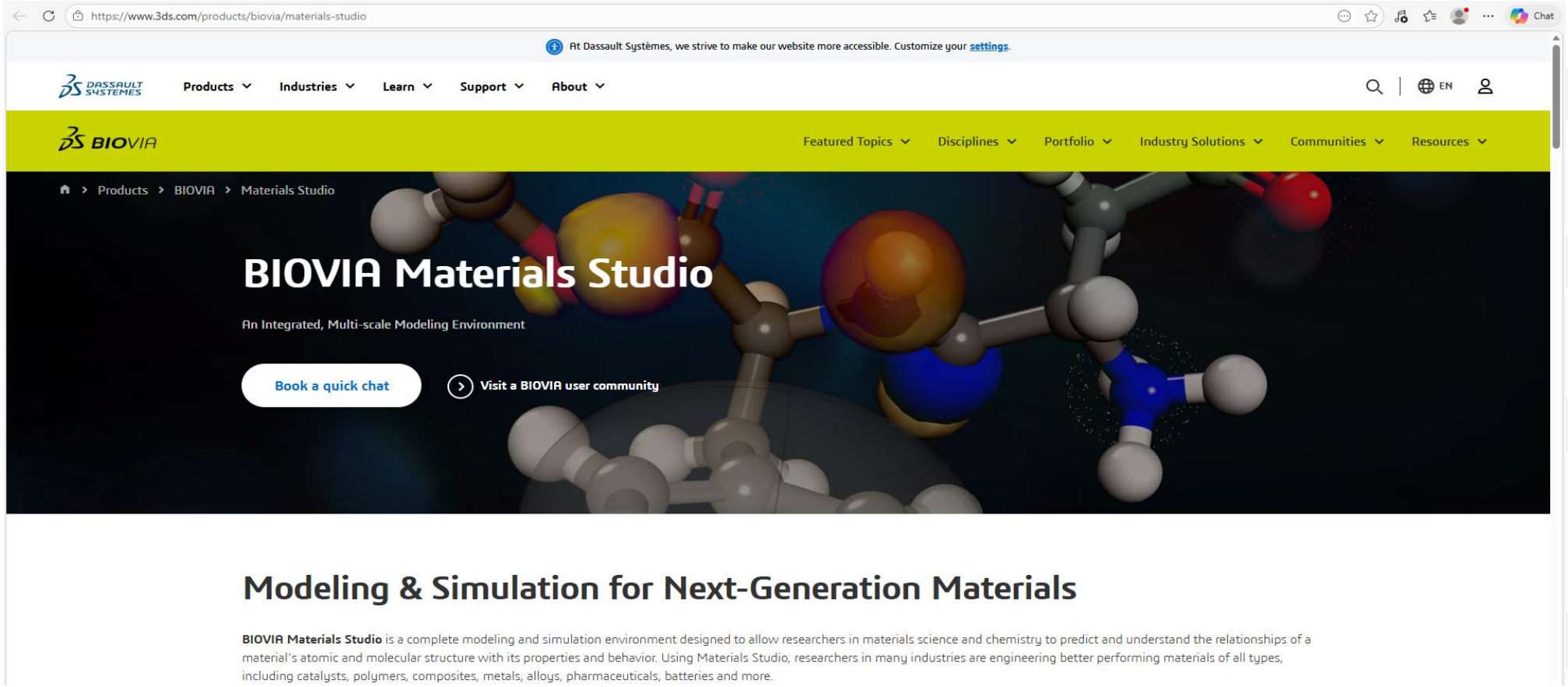
II. How to log onto the HPC account?

III. Coding for Lattice vector and *k* point in Reciprocal Space

I. Materials Studio

Software for simulating and modeling materials

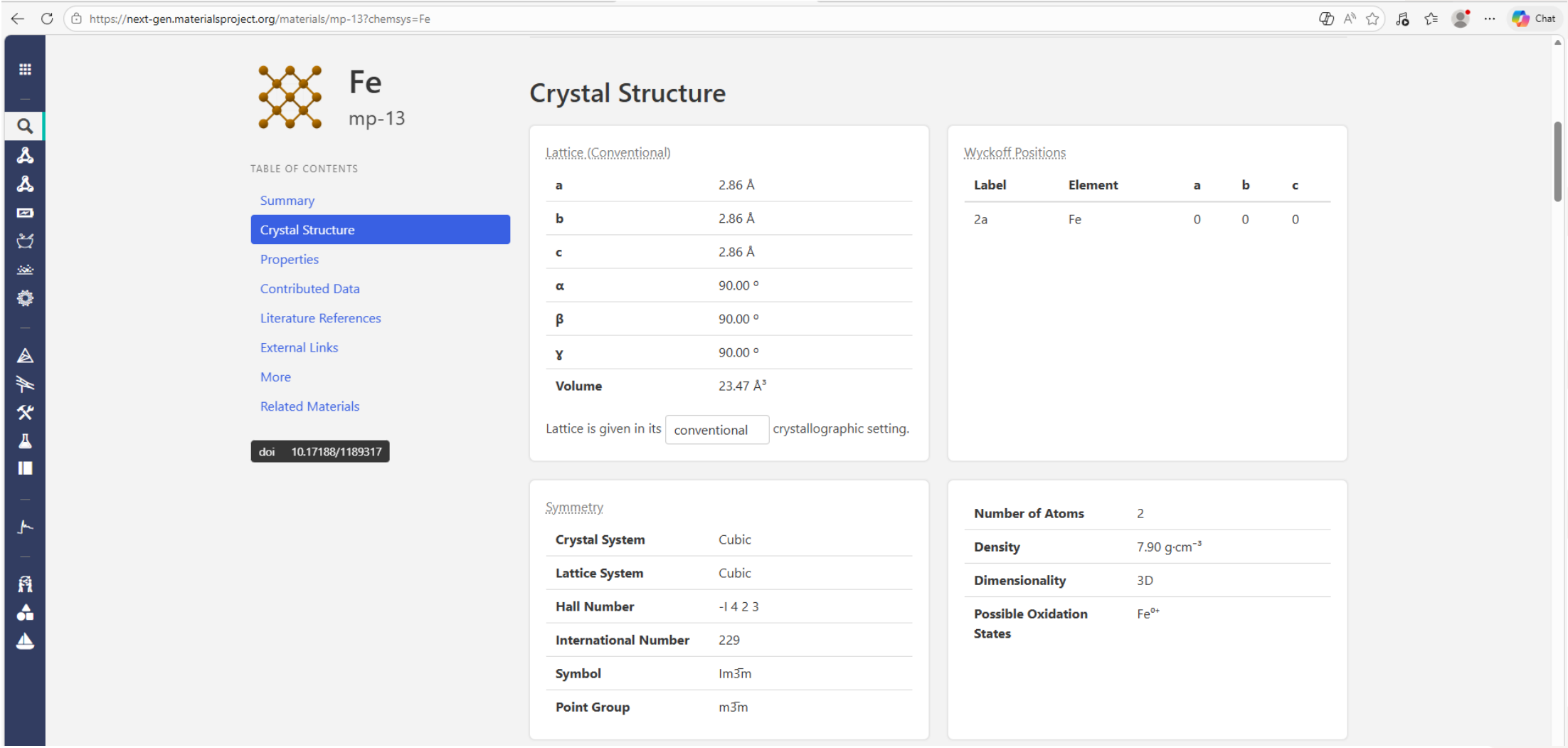
Developed and distributed by BIOVIA (formerly Accelrys), a firm specializing in research software for computational chemistry, bioinformatics, cheminformatics, molecular dynamics simulation, and quantum mechanics



https://www.3ds.com/products/biovia/materials-studio

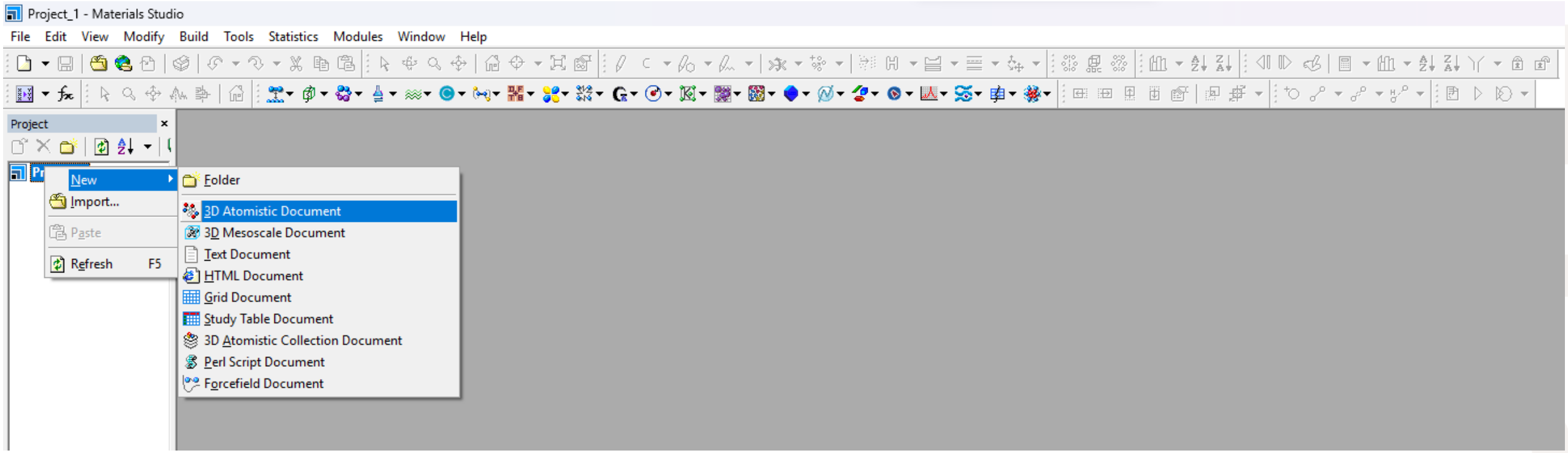
I. Materials Studio

From Materials Project, obtain the space group, lattice constant and Wyckoff position of each atom



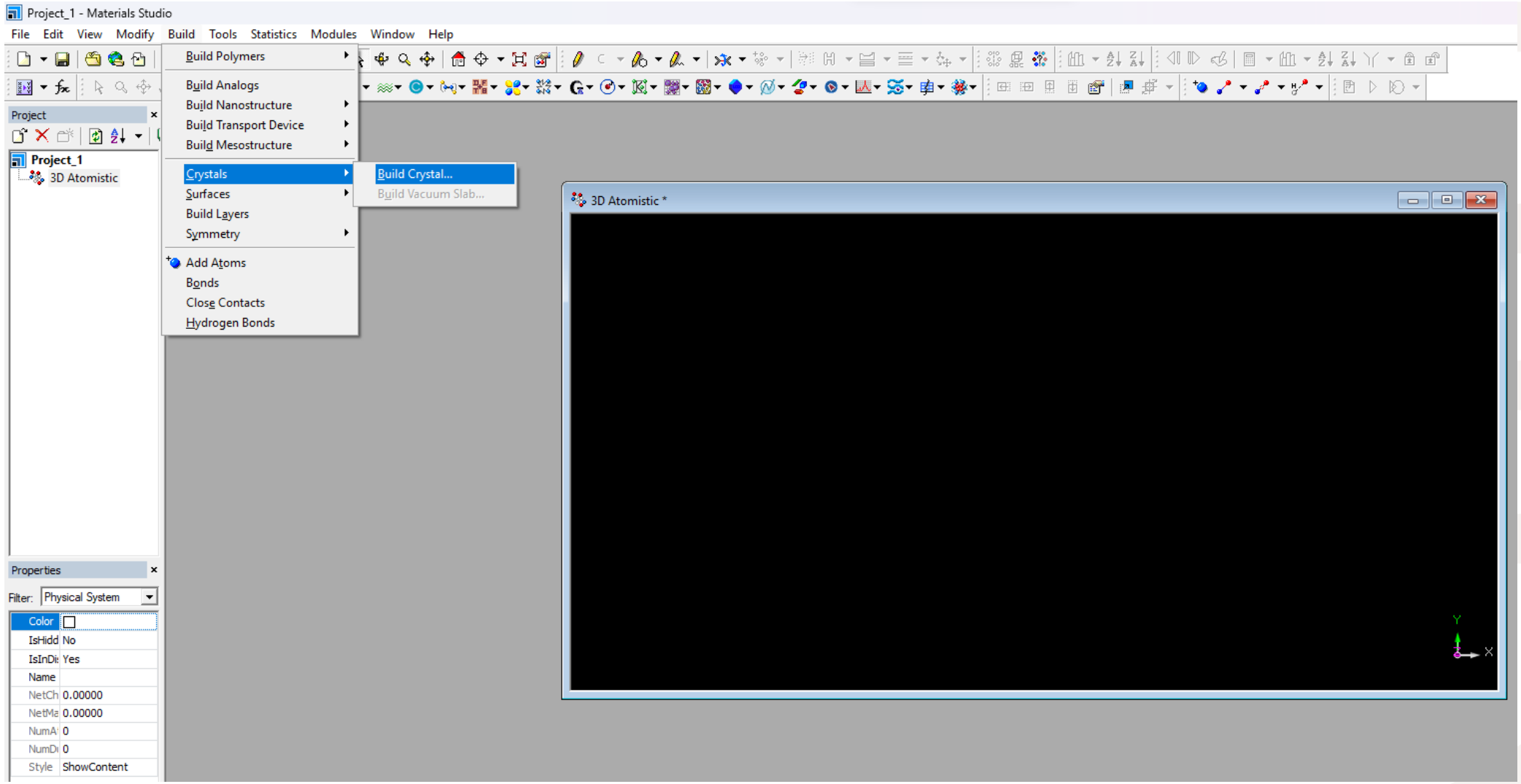
I. Materials Studio

Step One: Create a new 3D Atomistic Document in Materials Studio



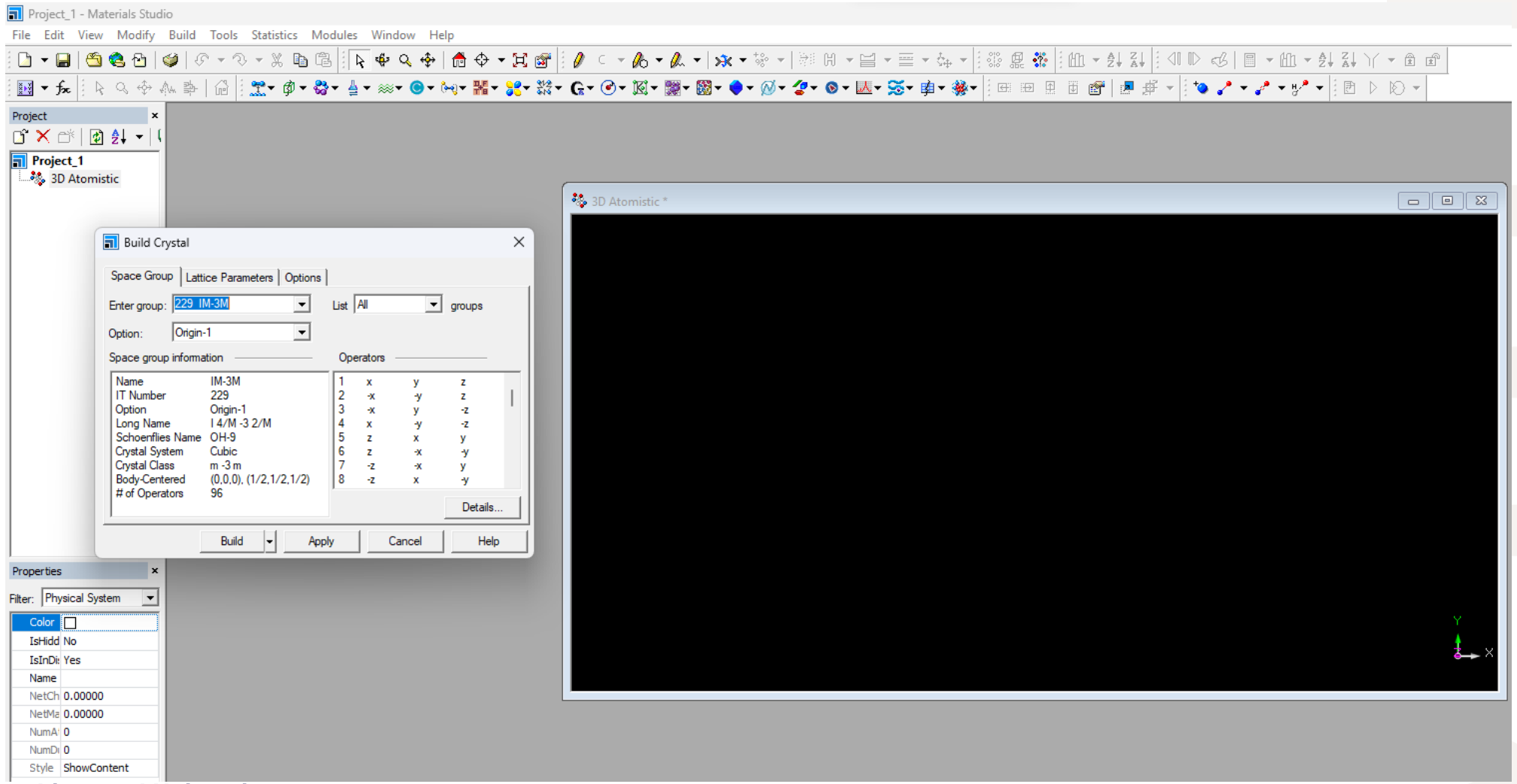
I. Materials Studio

Step Two: Build Crystal



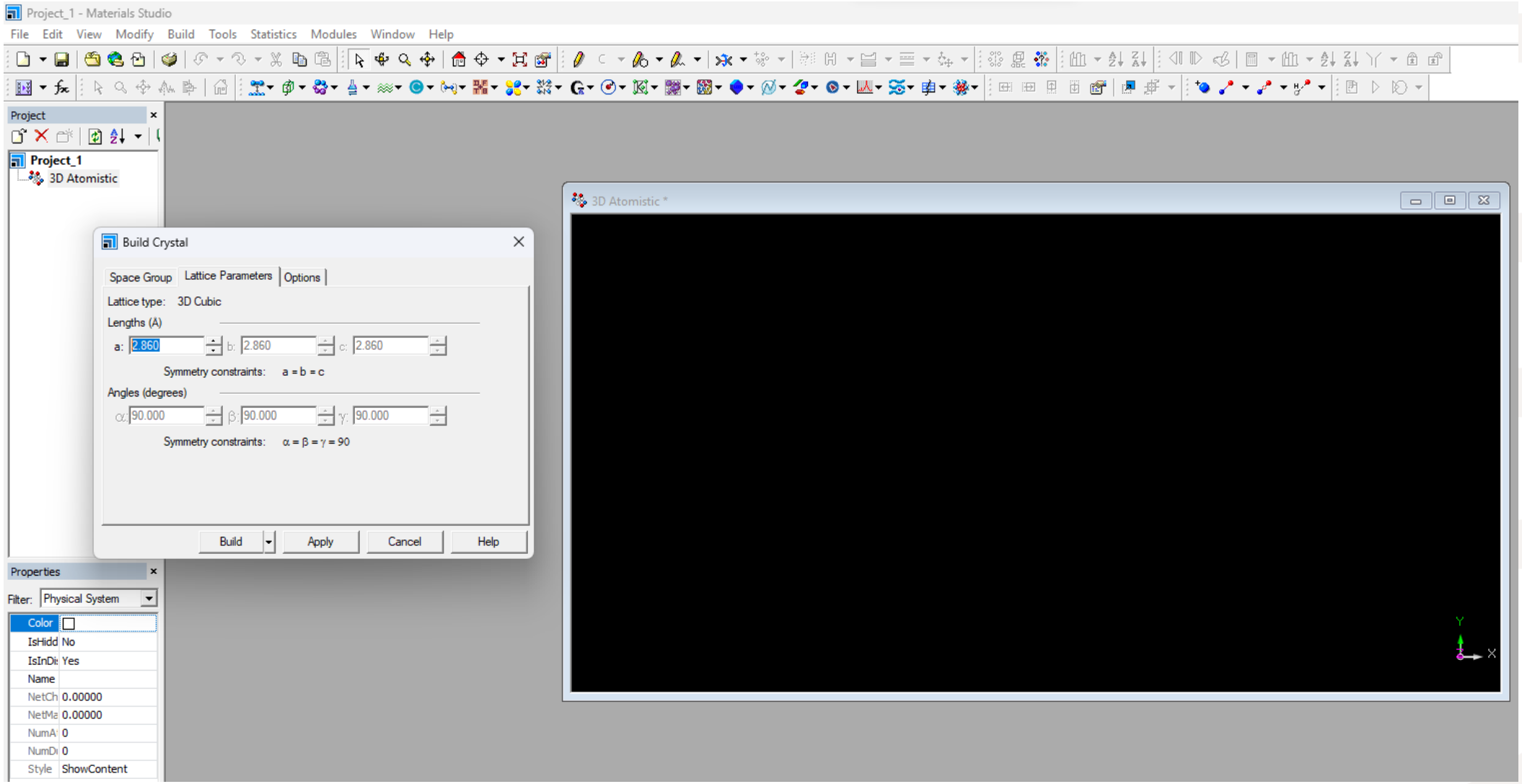
I. Materials Studio

Step Three: Enter the Space Group Number



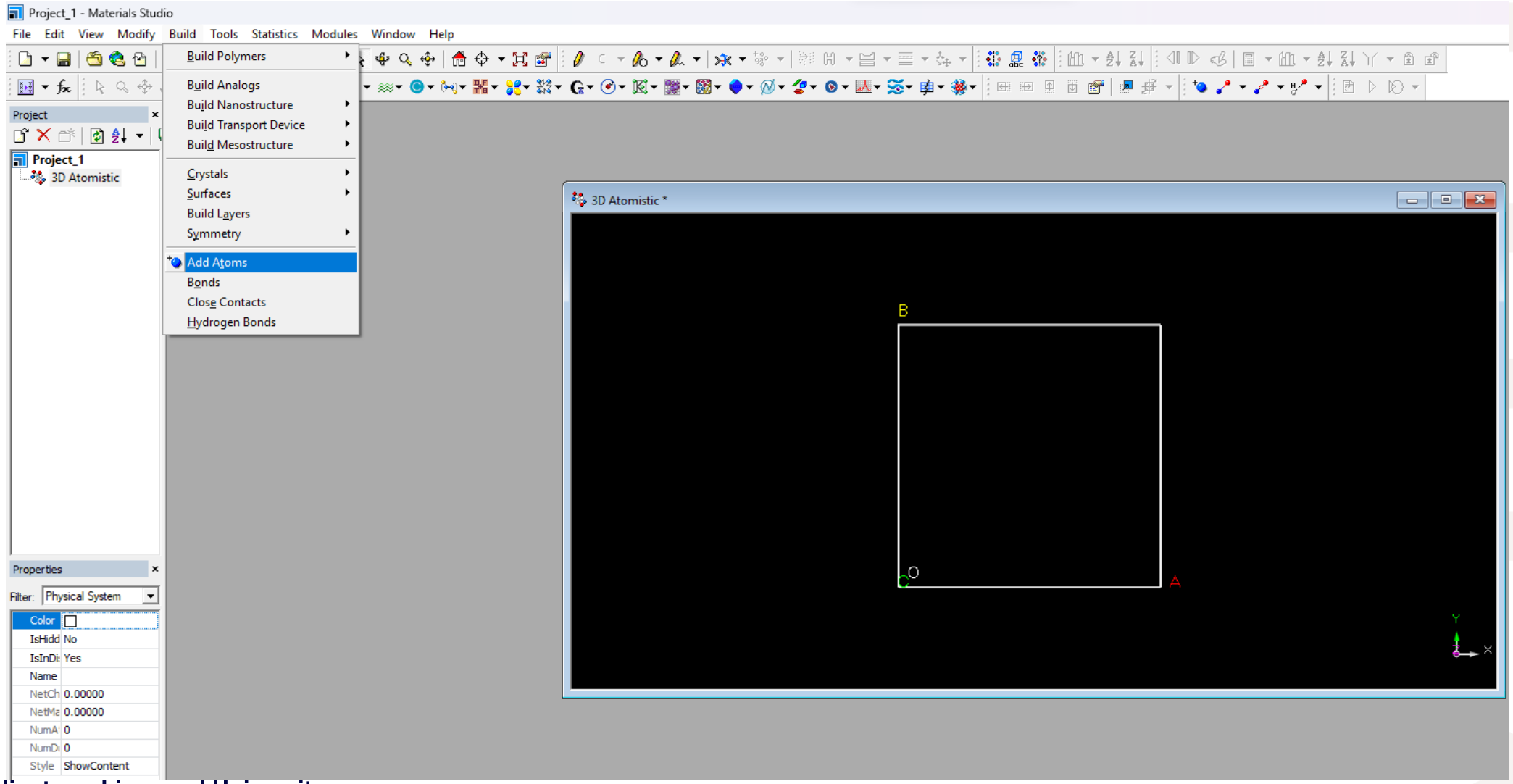
I. Materials Studio

Step Four: Enter the Lattice Constant

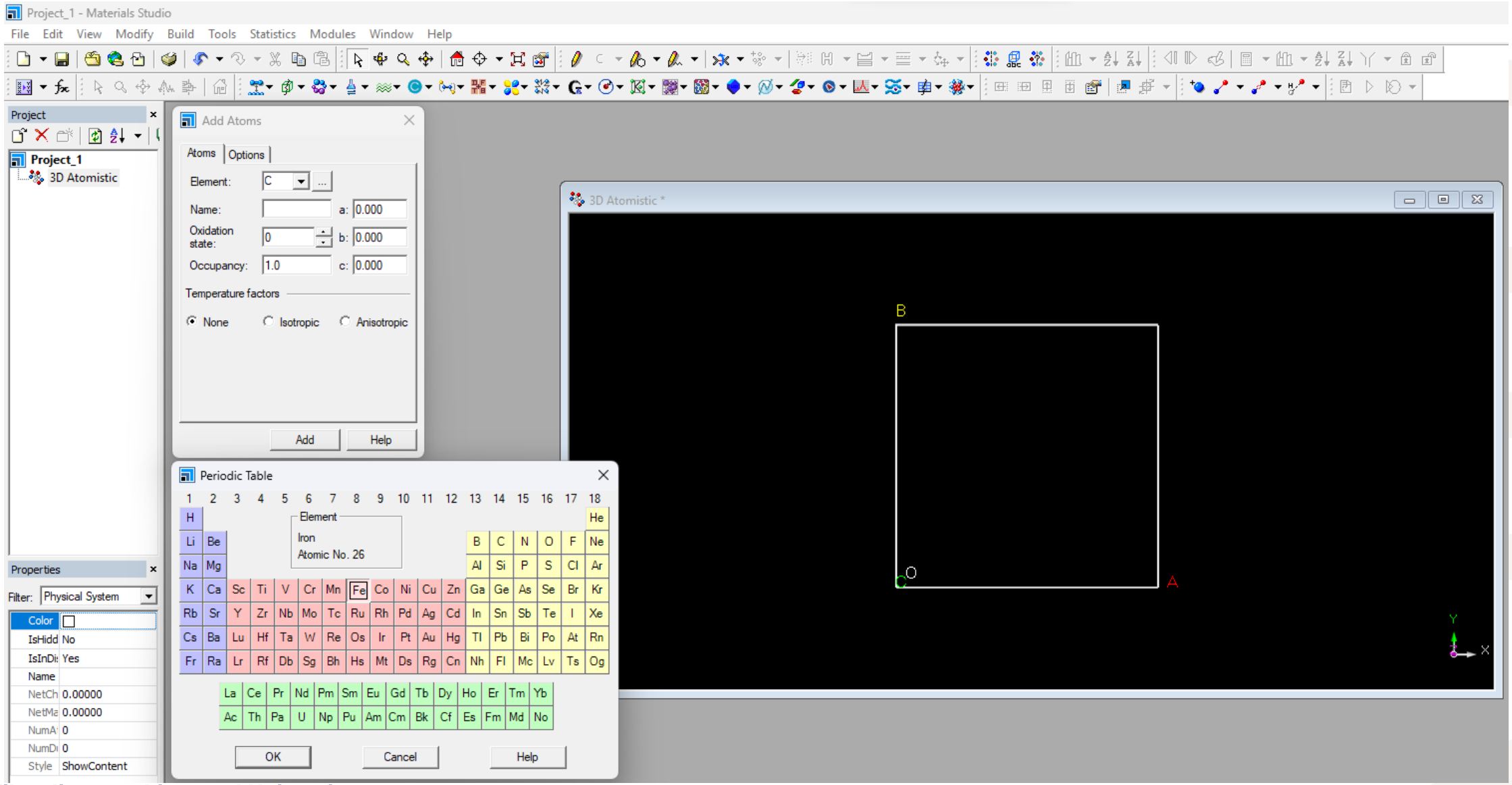


I. Materials Studio

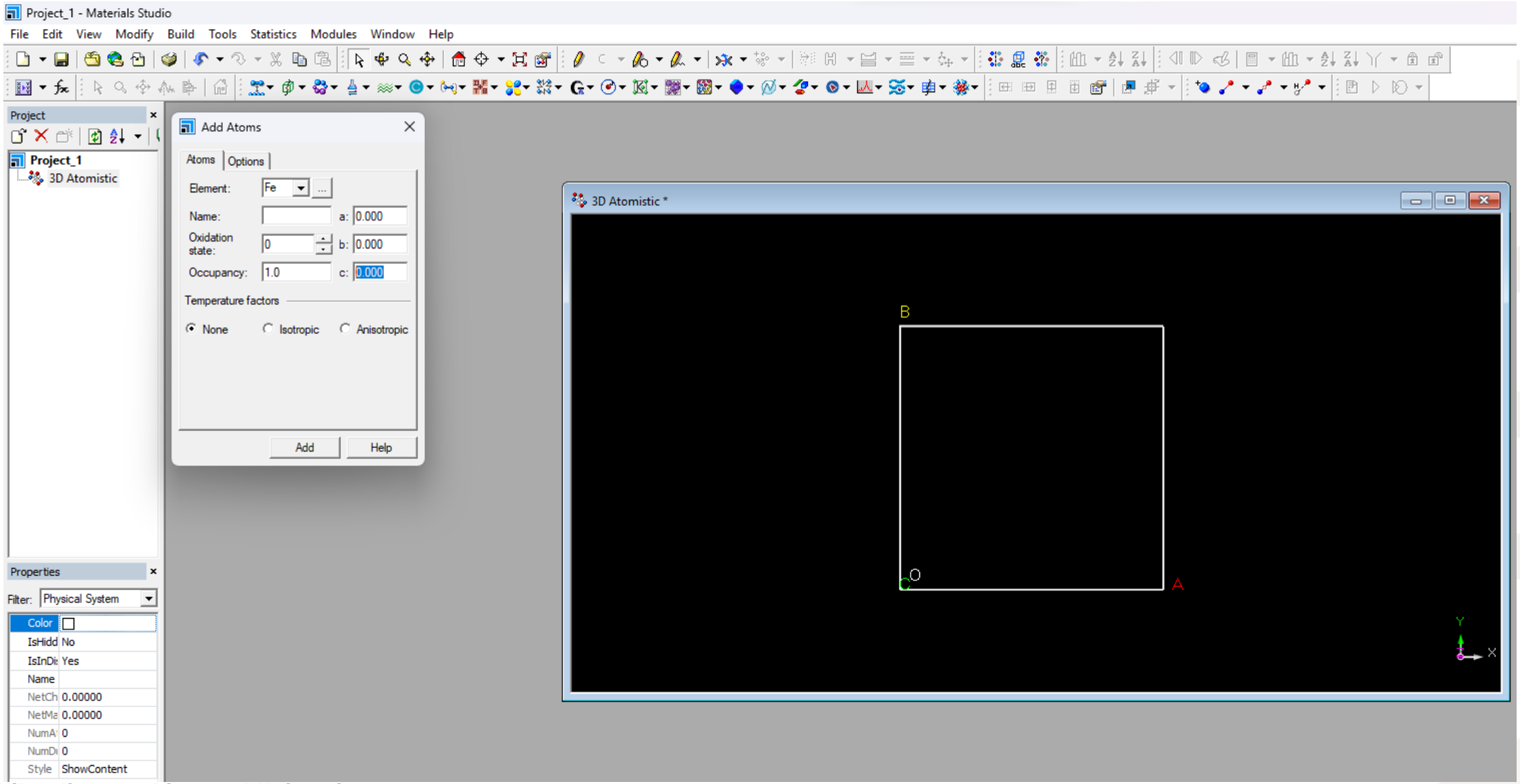
Step Five: Add atom coordinate



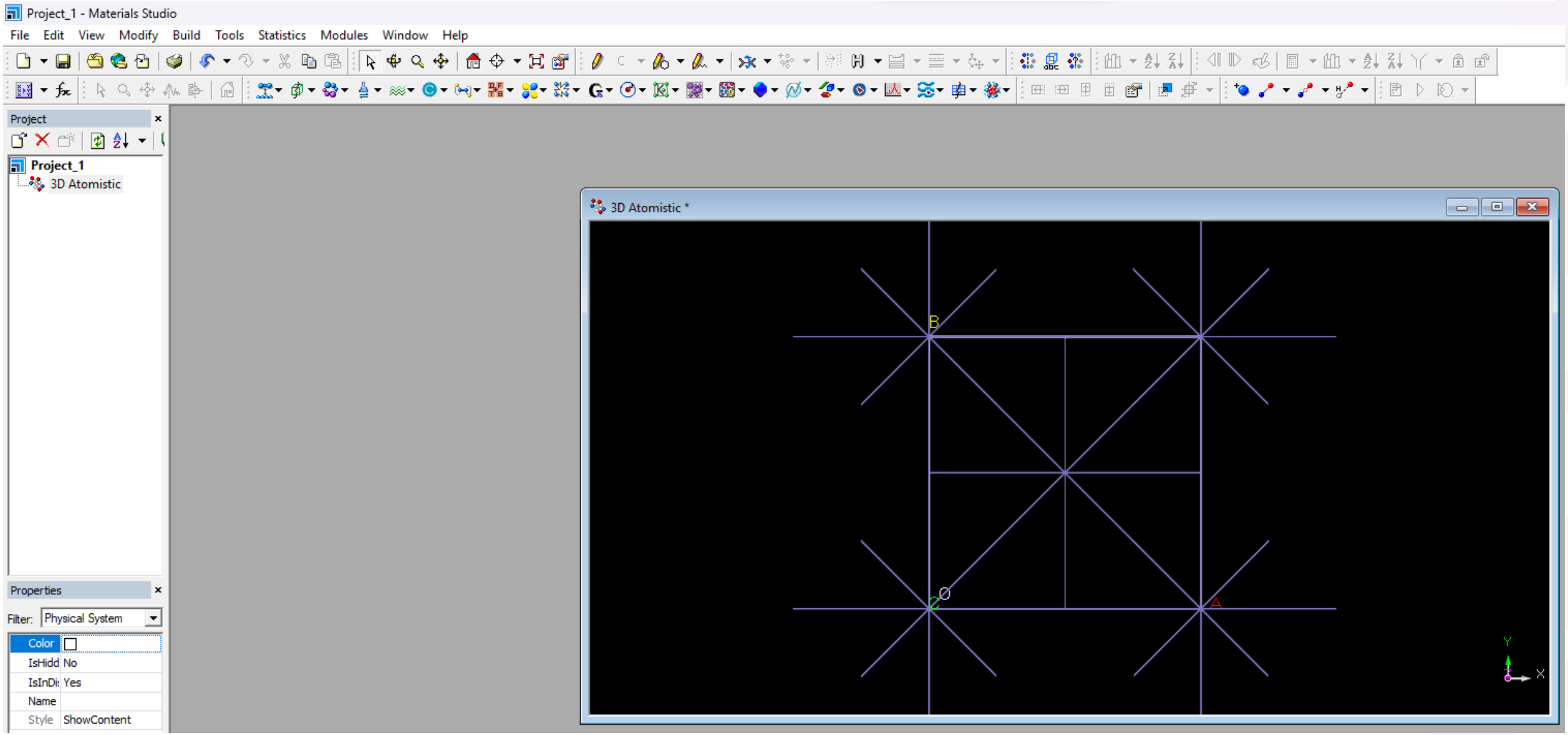
I. Materials Studio



I. Materials Studio

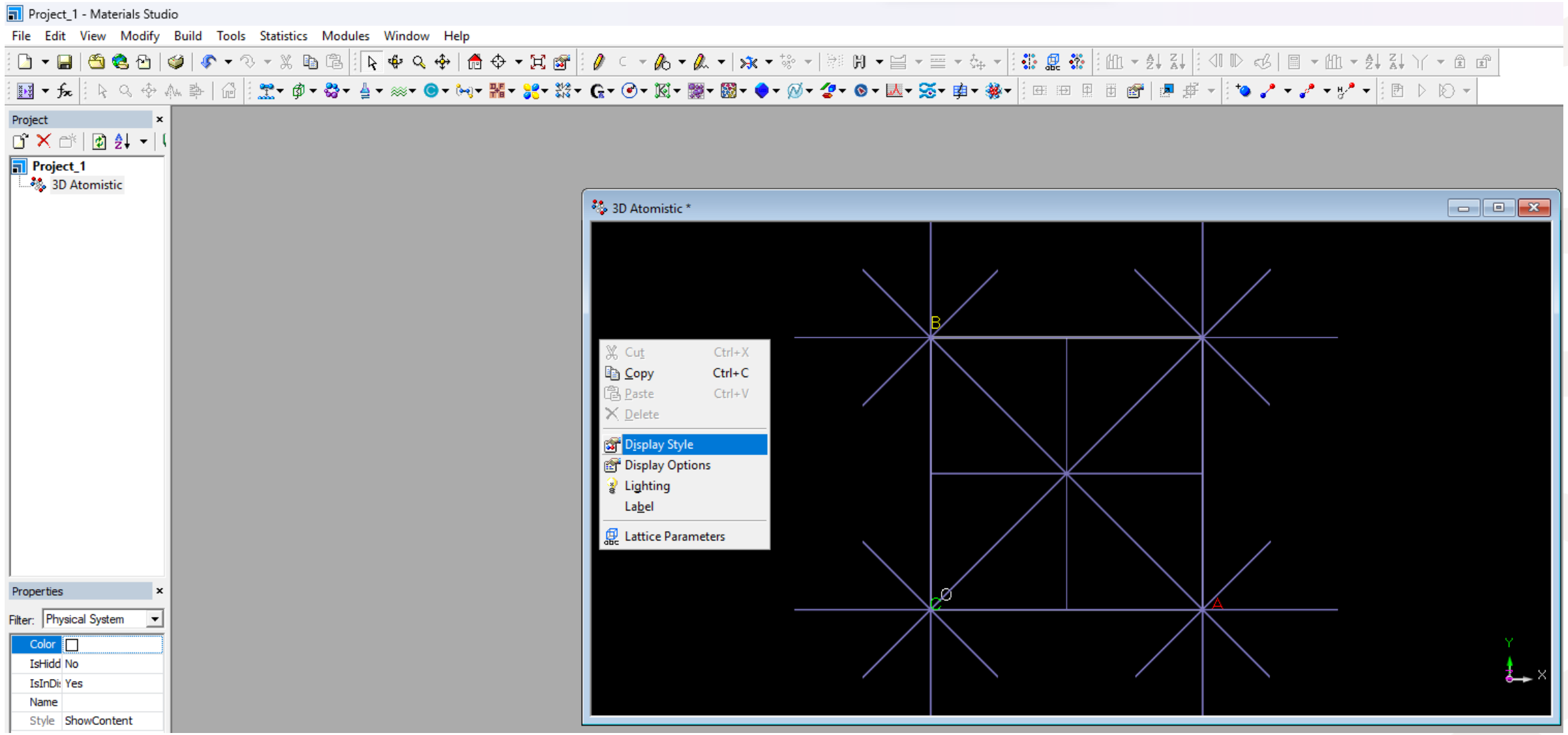


I. Materials Studio

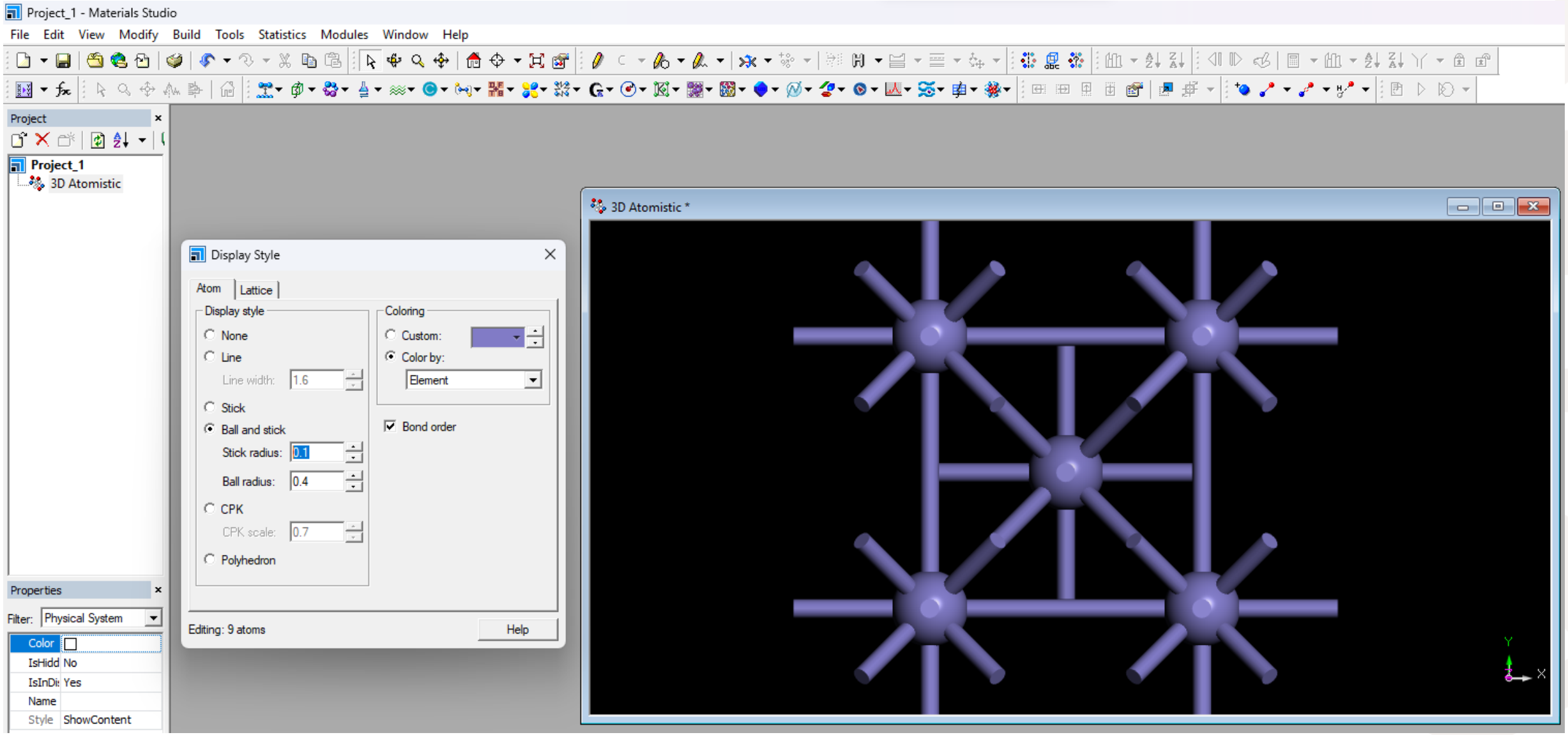


I. Materials Studio

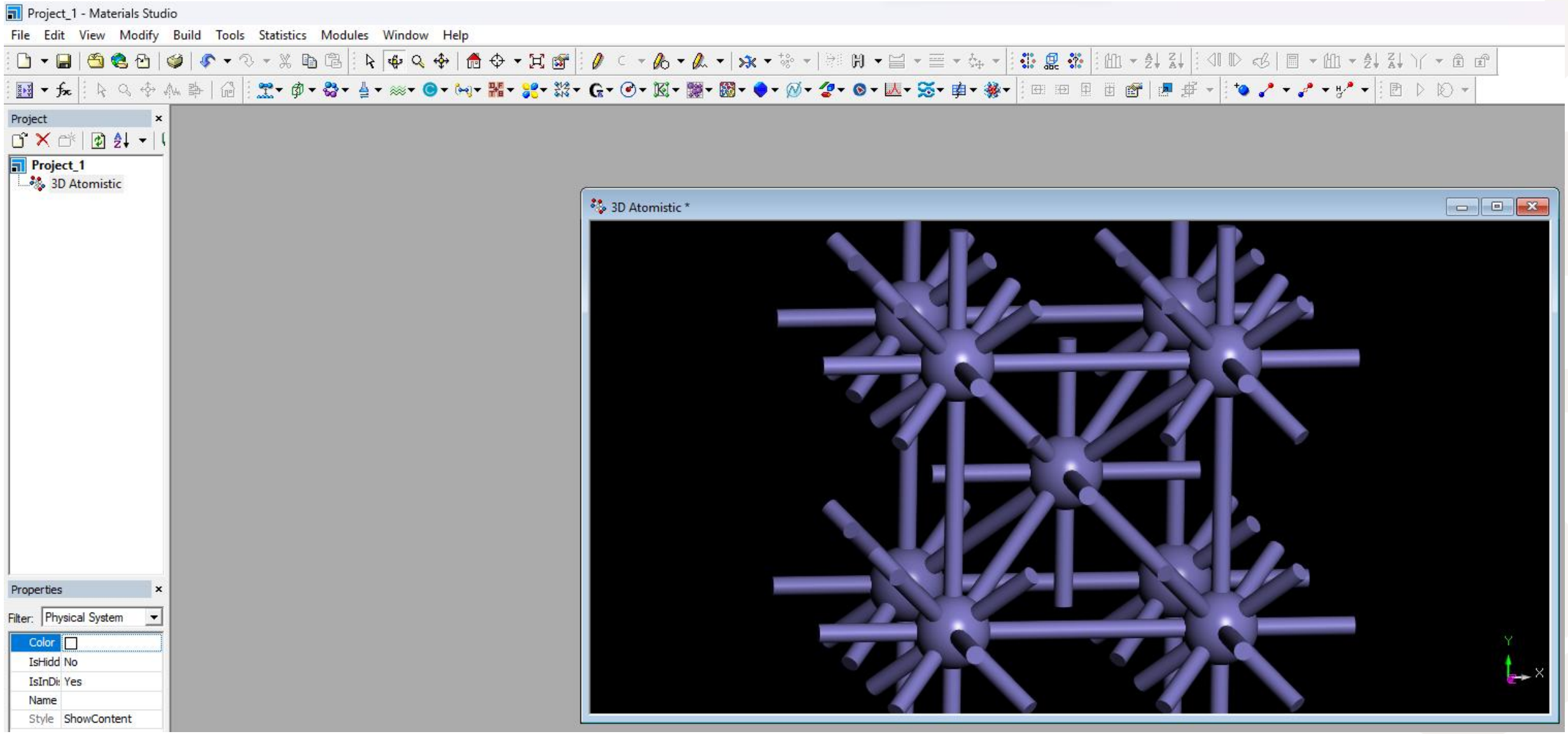
Step Six: Adjust the Display Style



I. Materials Studio

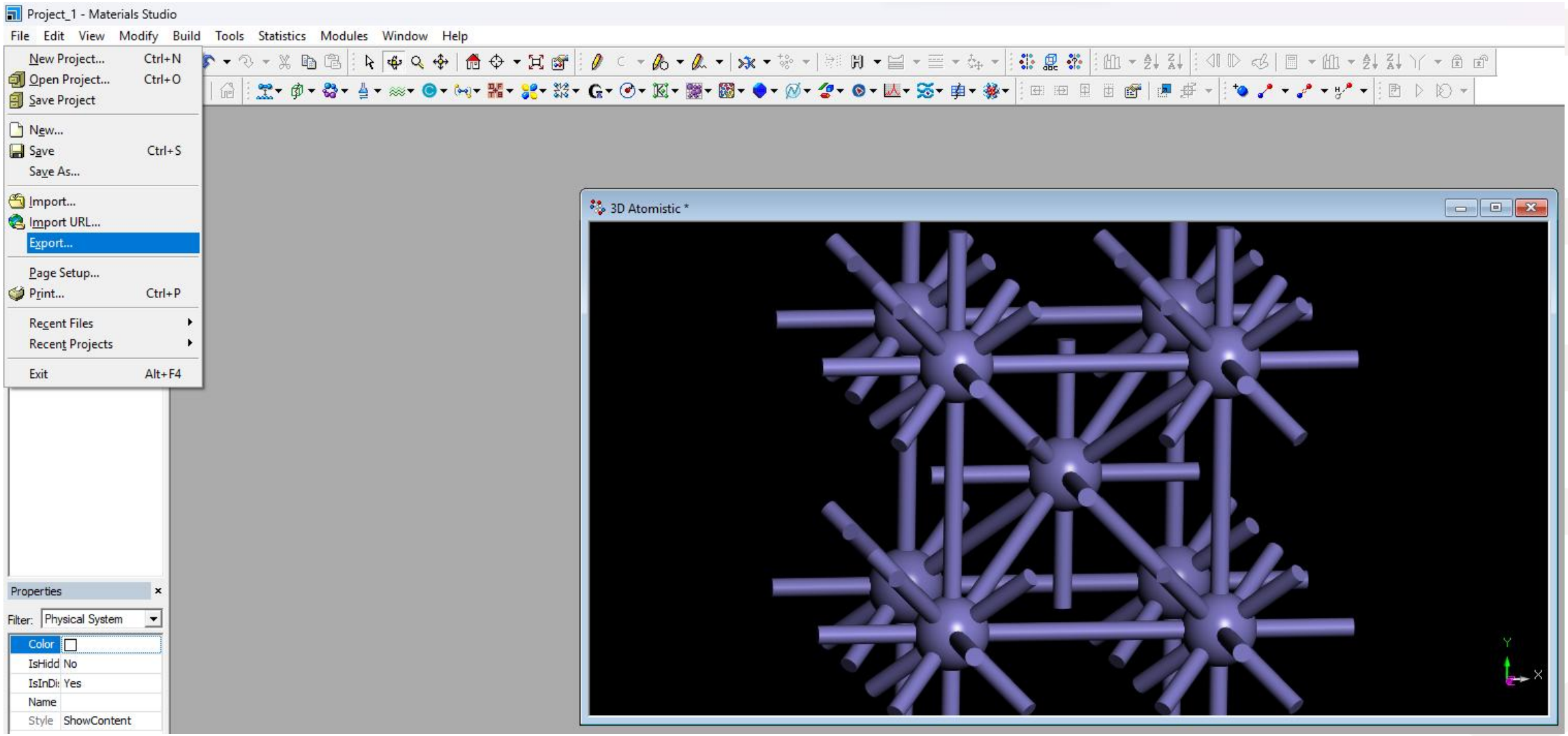


I. Materials Studio

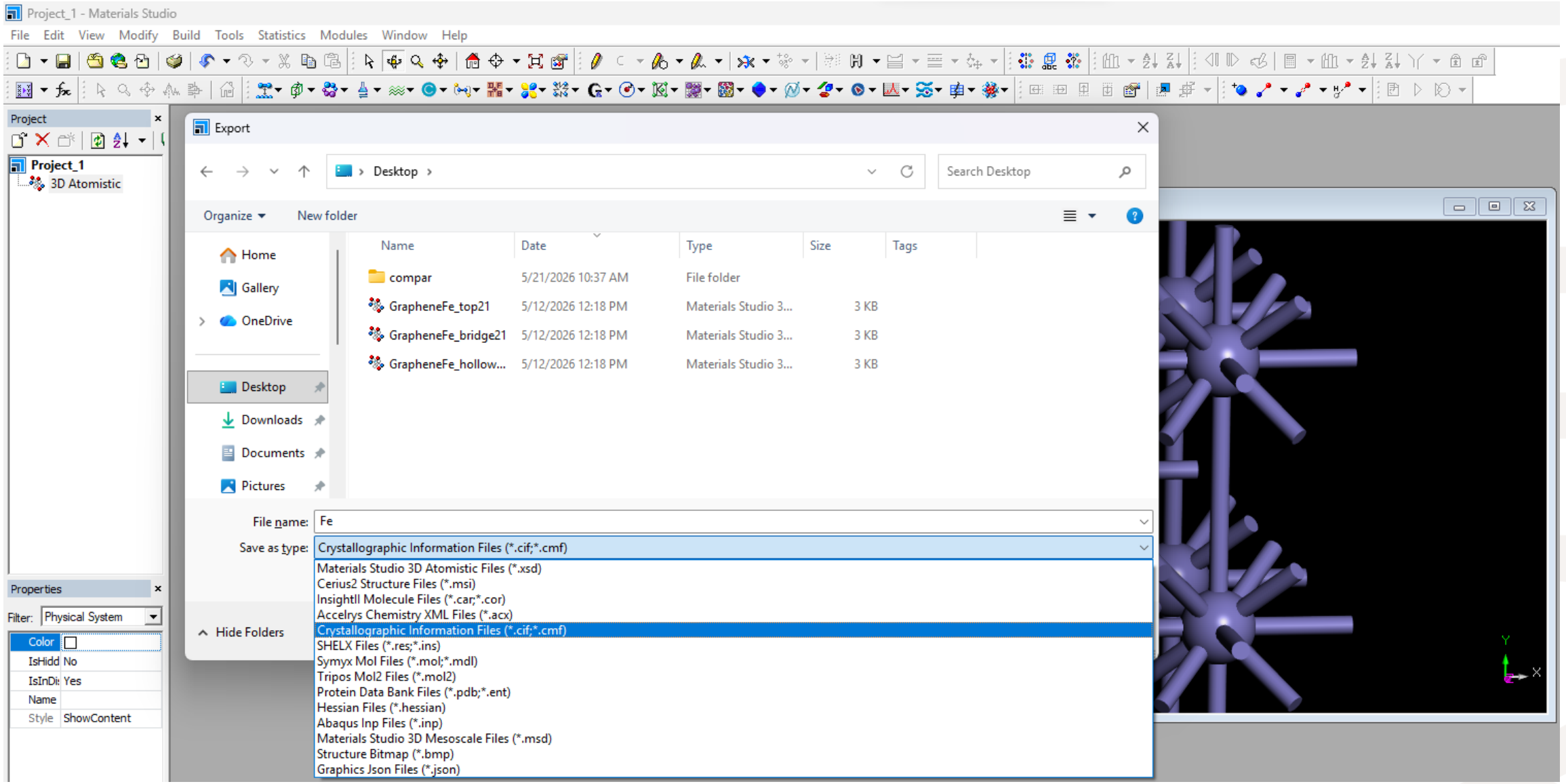


I. Materials Studio

Step Seven: Export crystal in the ' .cif ' format

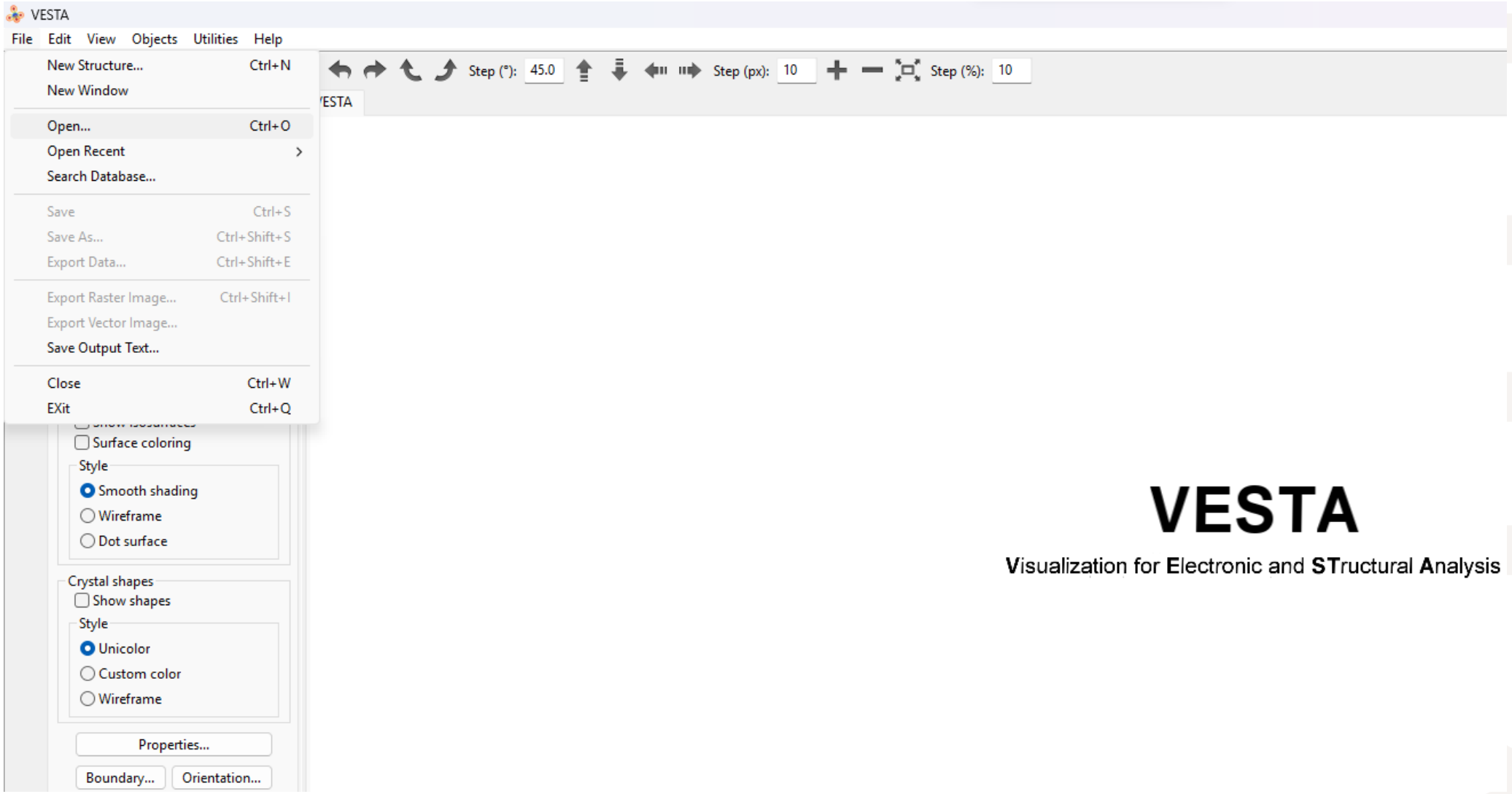


I. Materials Studio

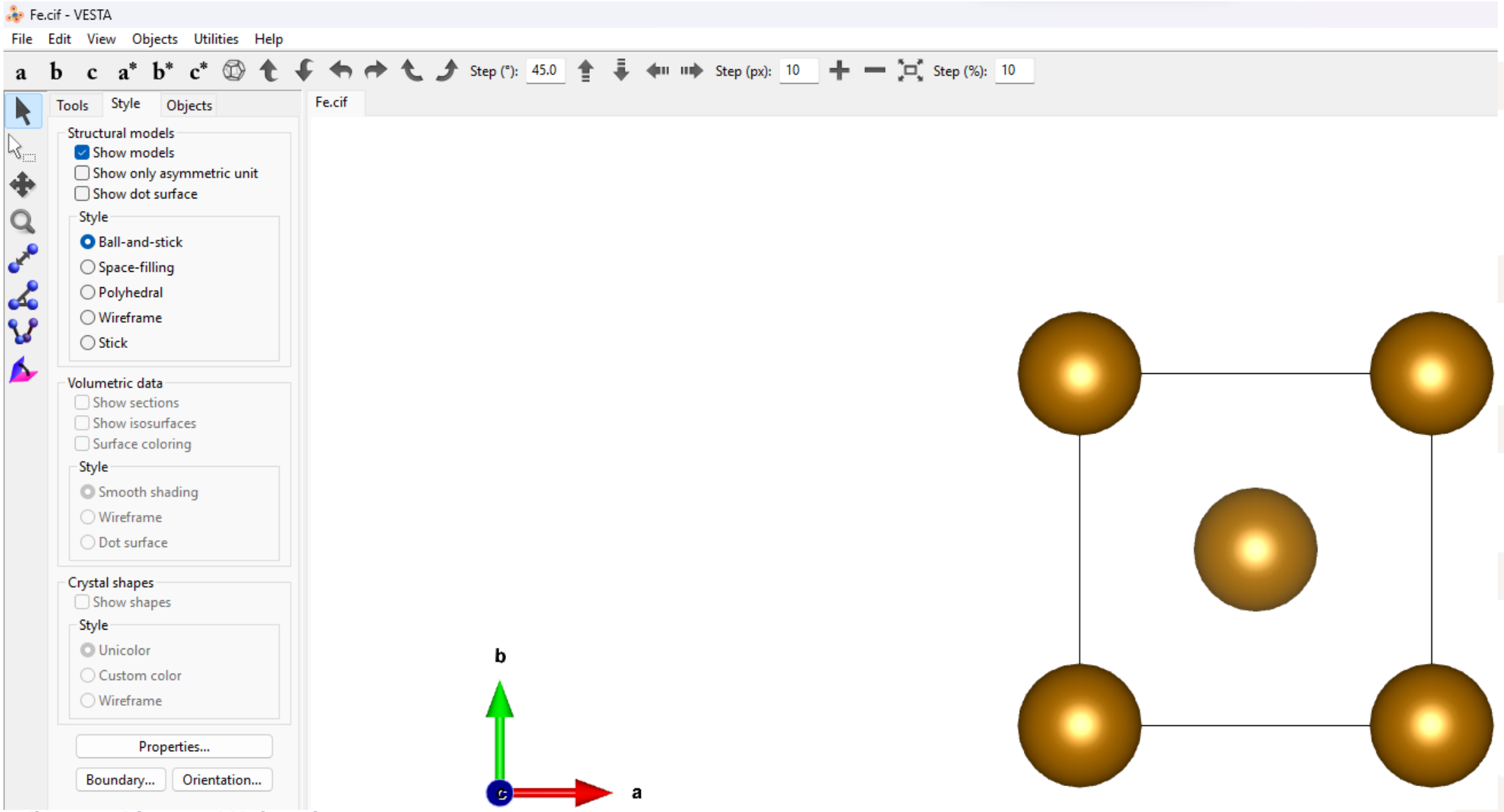


I. Materials Studio

Step Eight: Open this ' .cif ' file with VESTA software

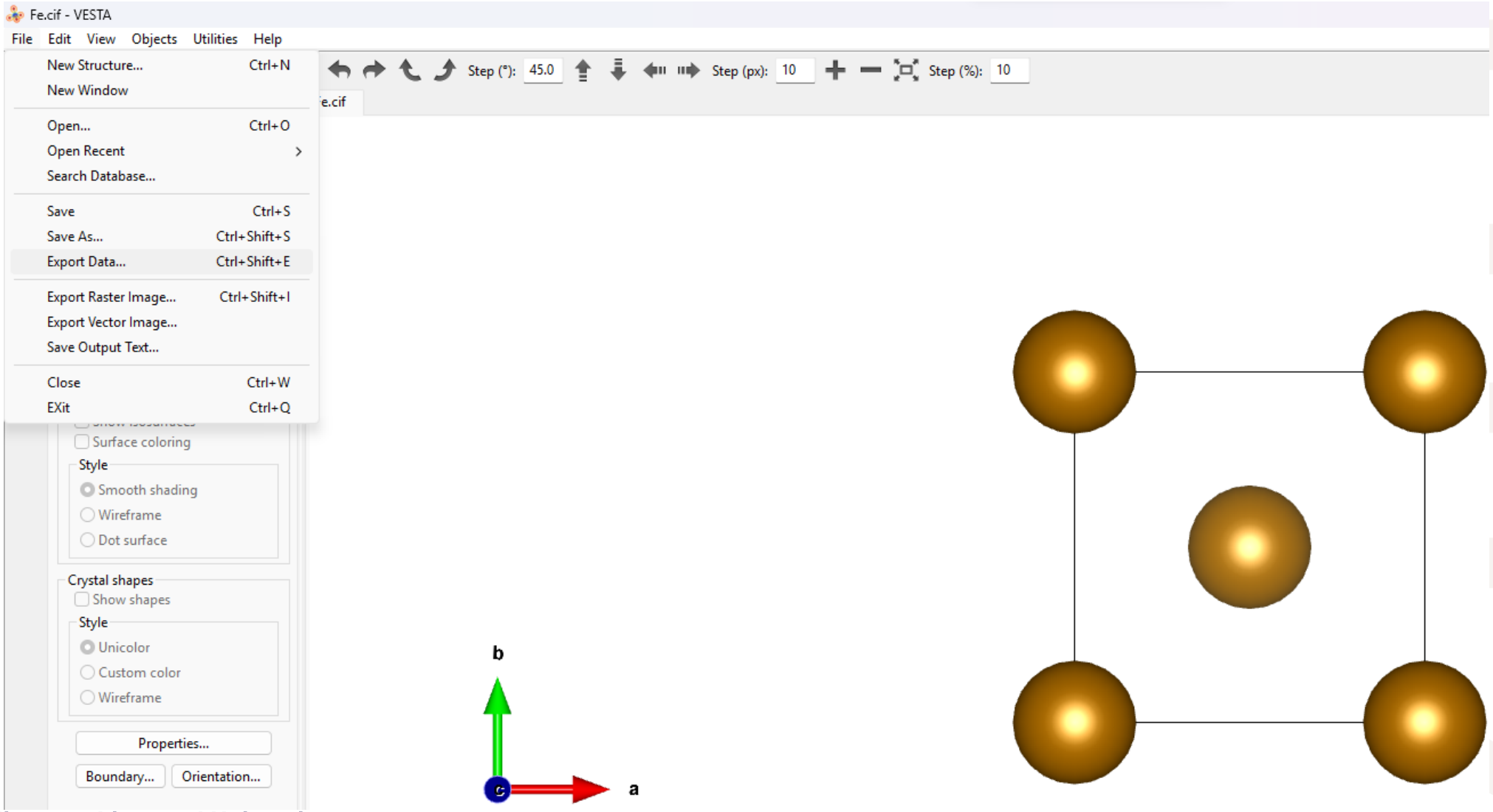


I. Materials Studio

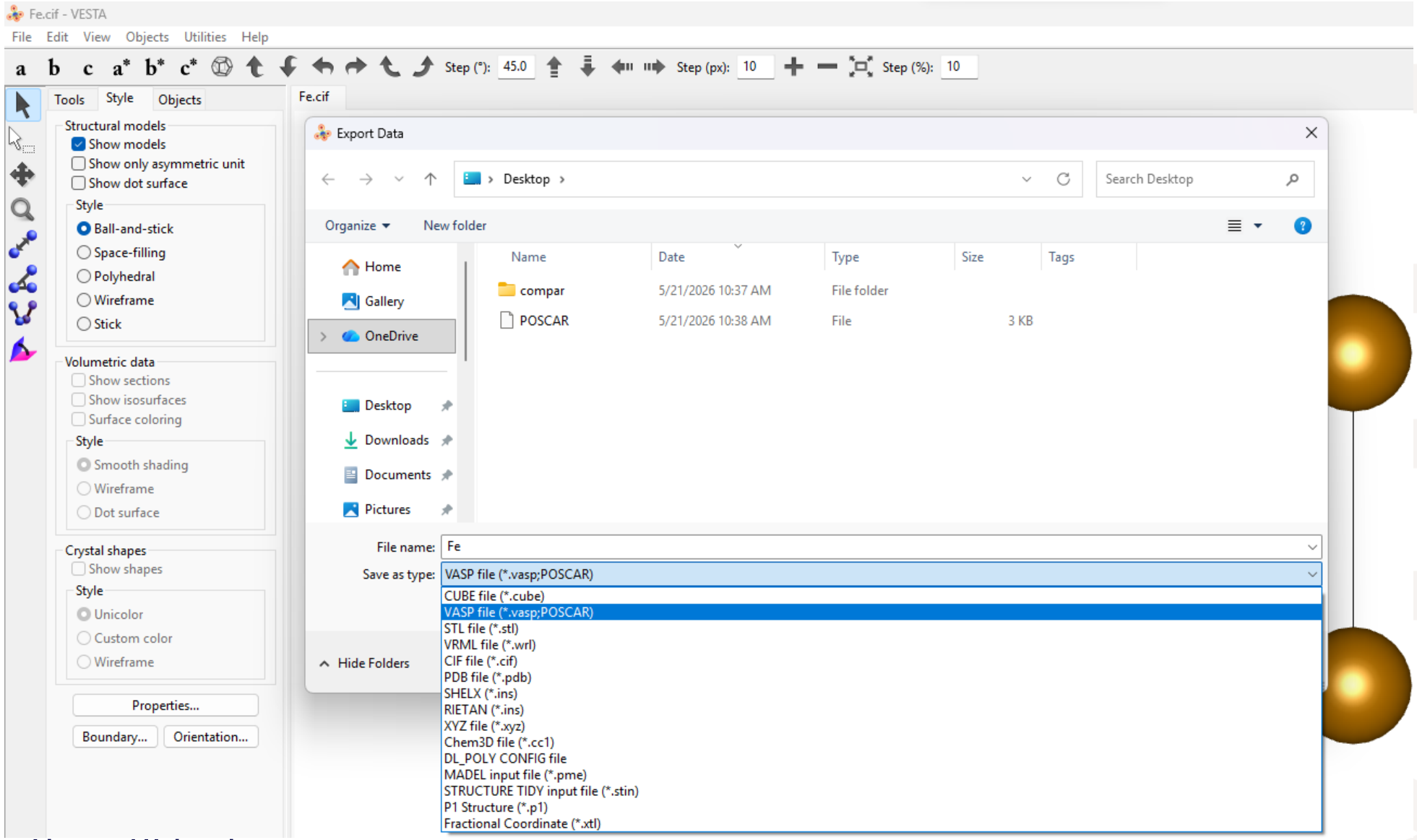


I. Materials Studio

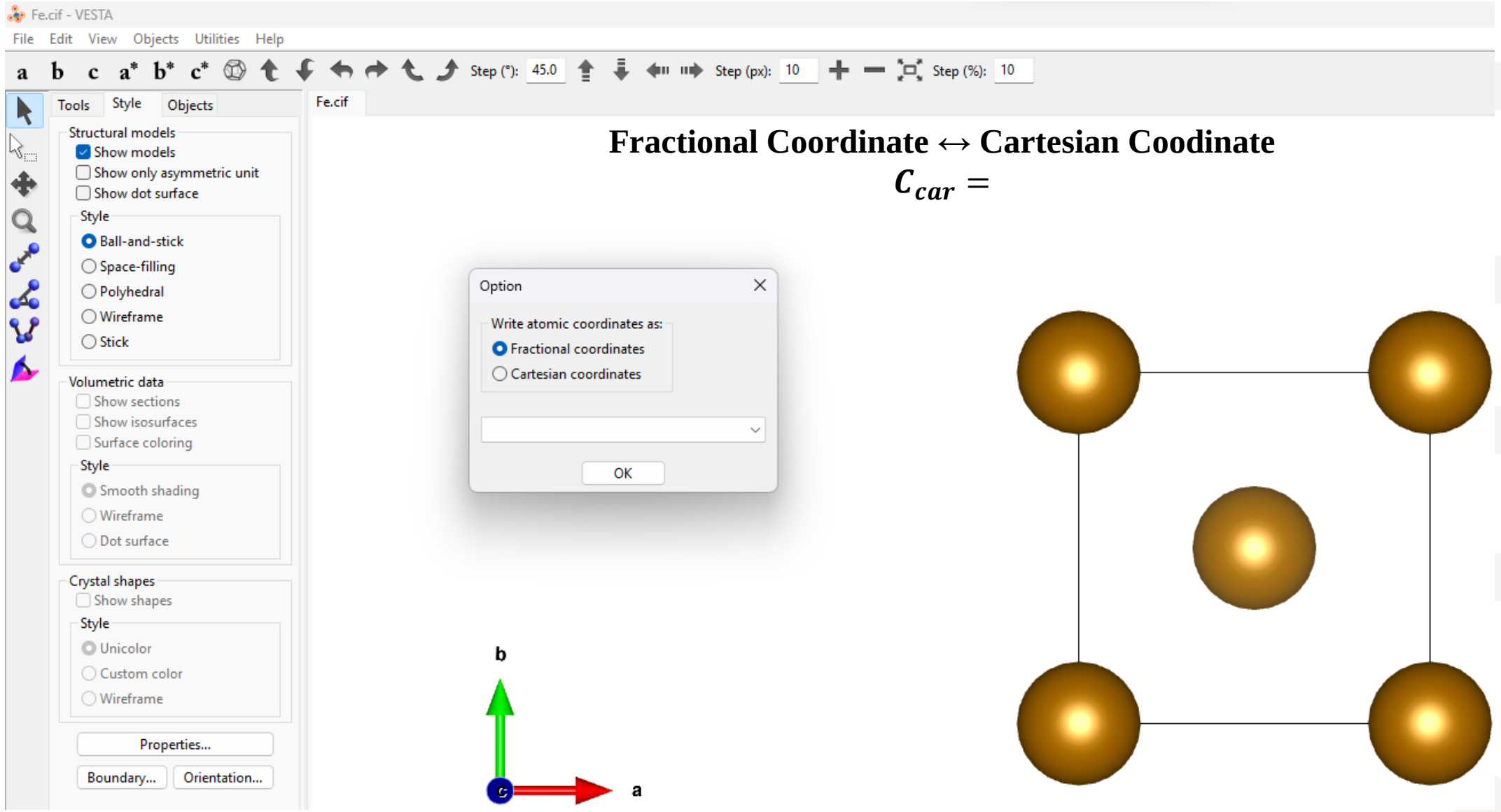
Step Nine: Export structure in 'POSCAR' format



I. Materials Studio



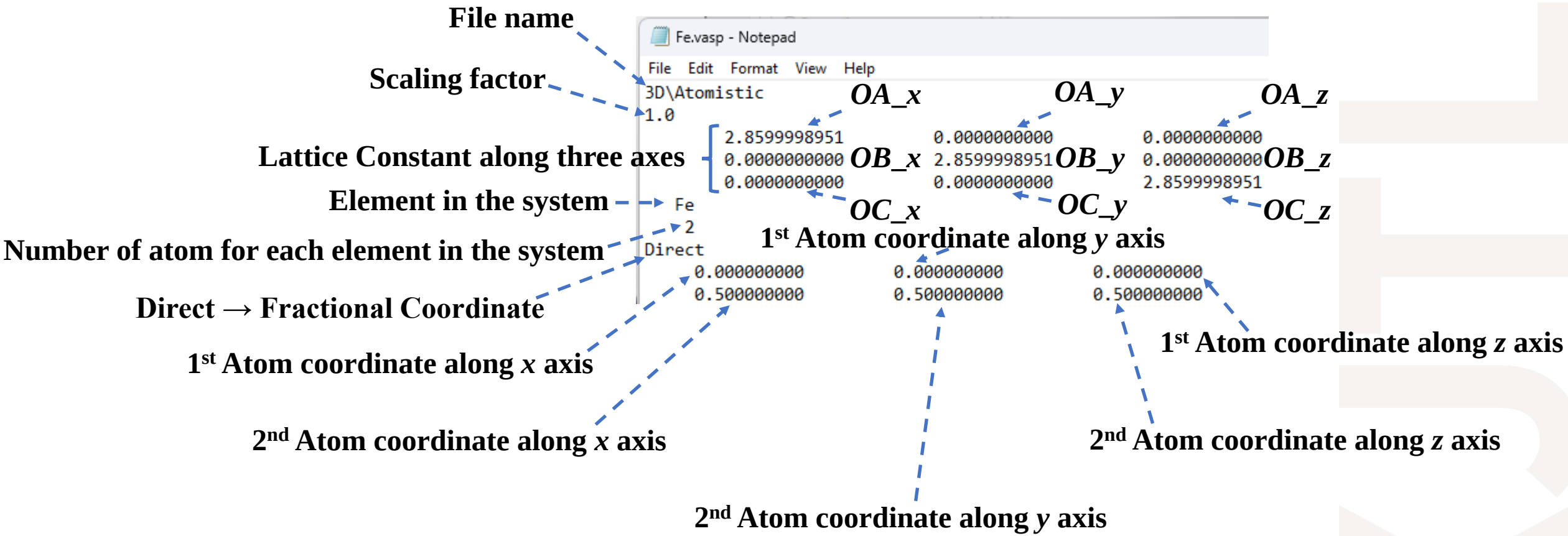
I. Materials Studio



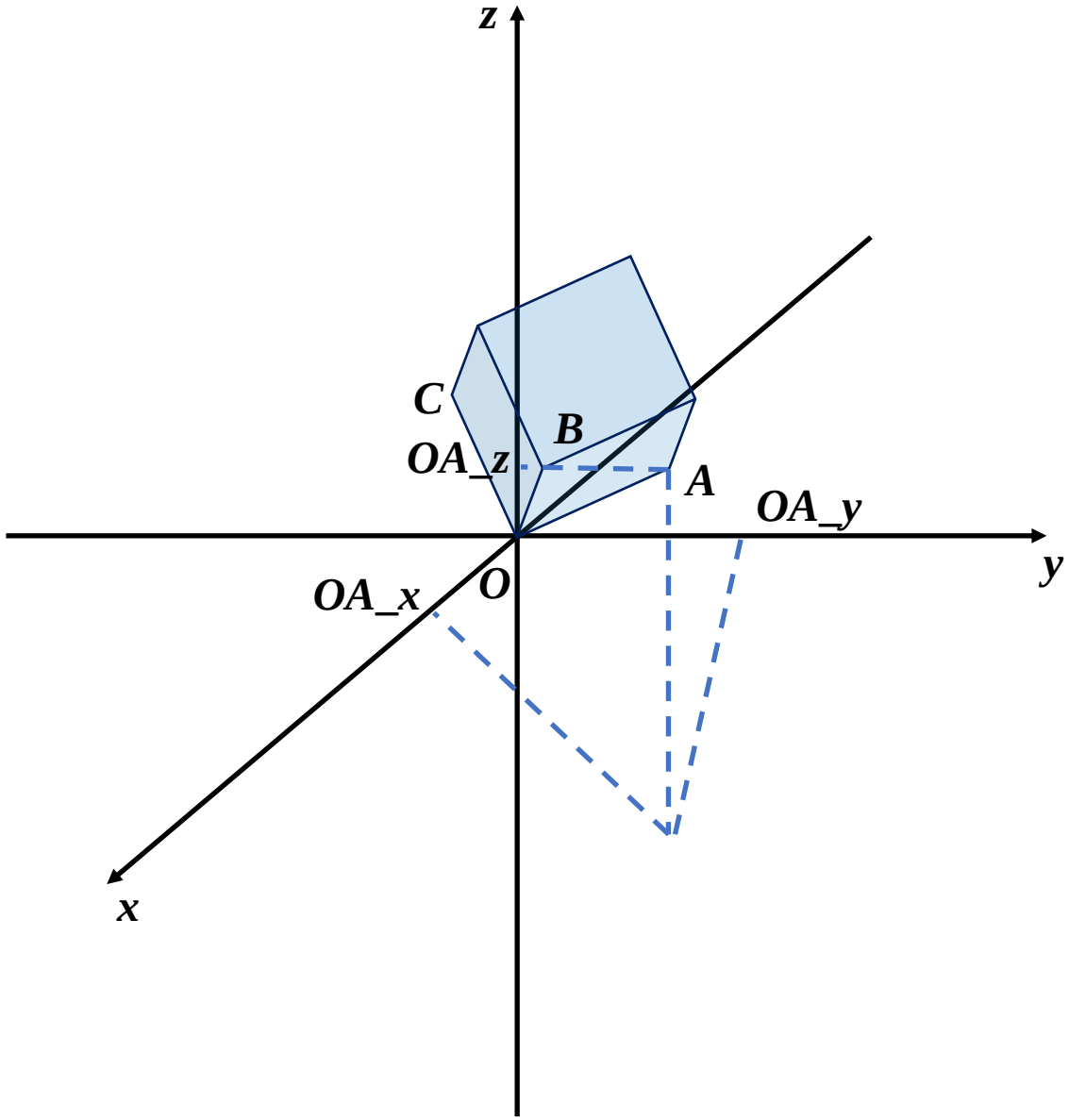
Fractional Coordinate ↔ Cartesian Coodinate

$$C_{car} =$$

I. Materials Studio



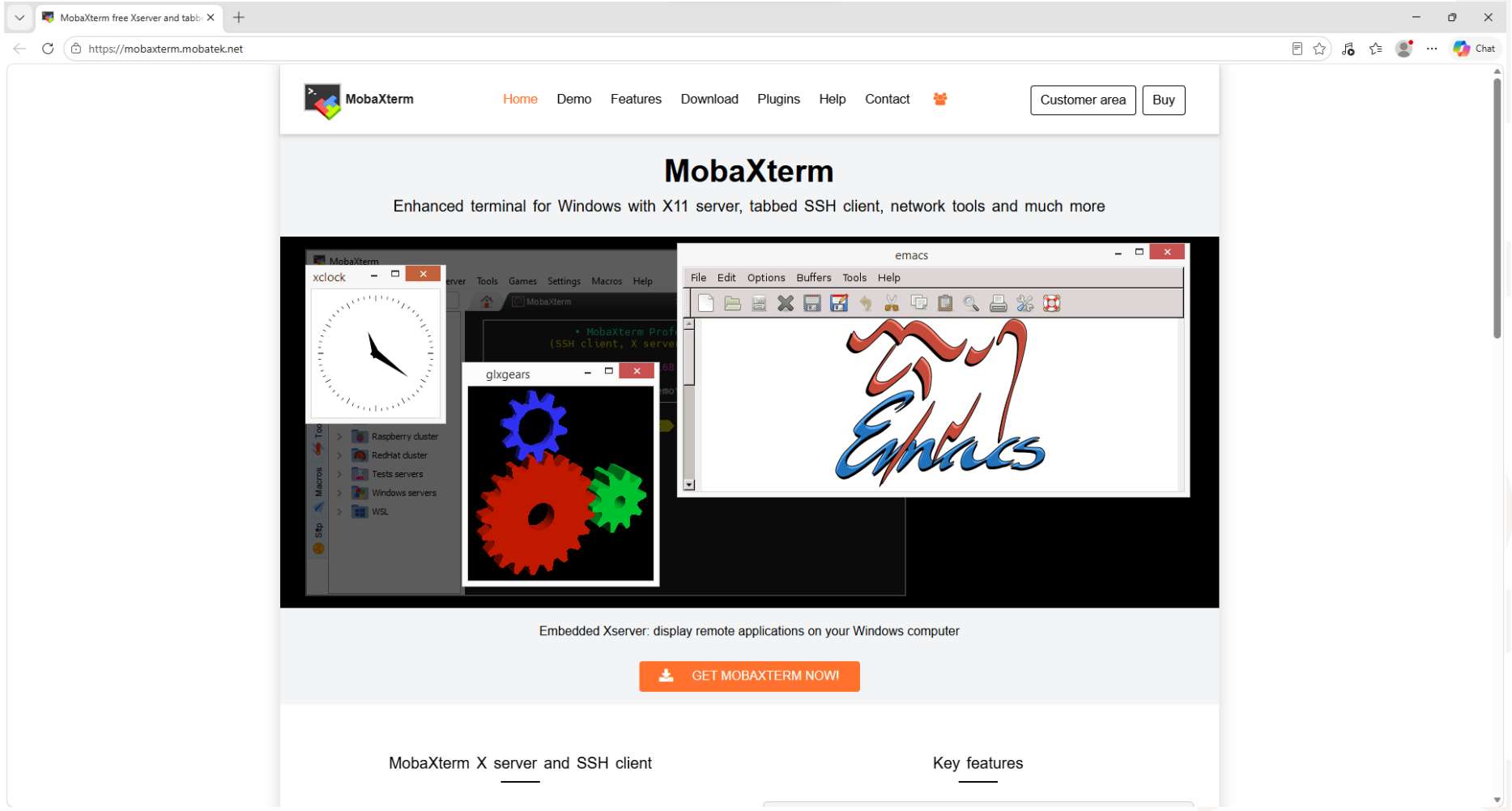
I. Materials Studio



II. How to log onto the HPC account?

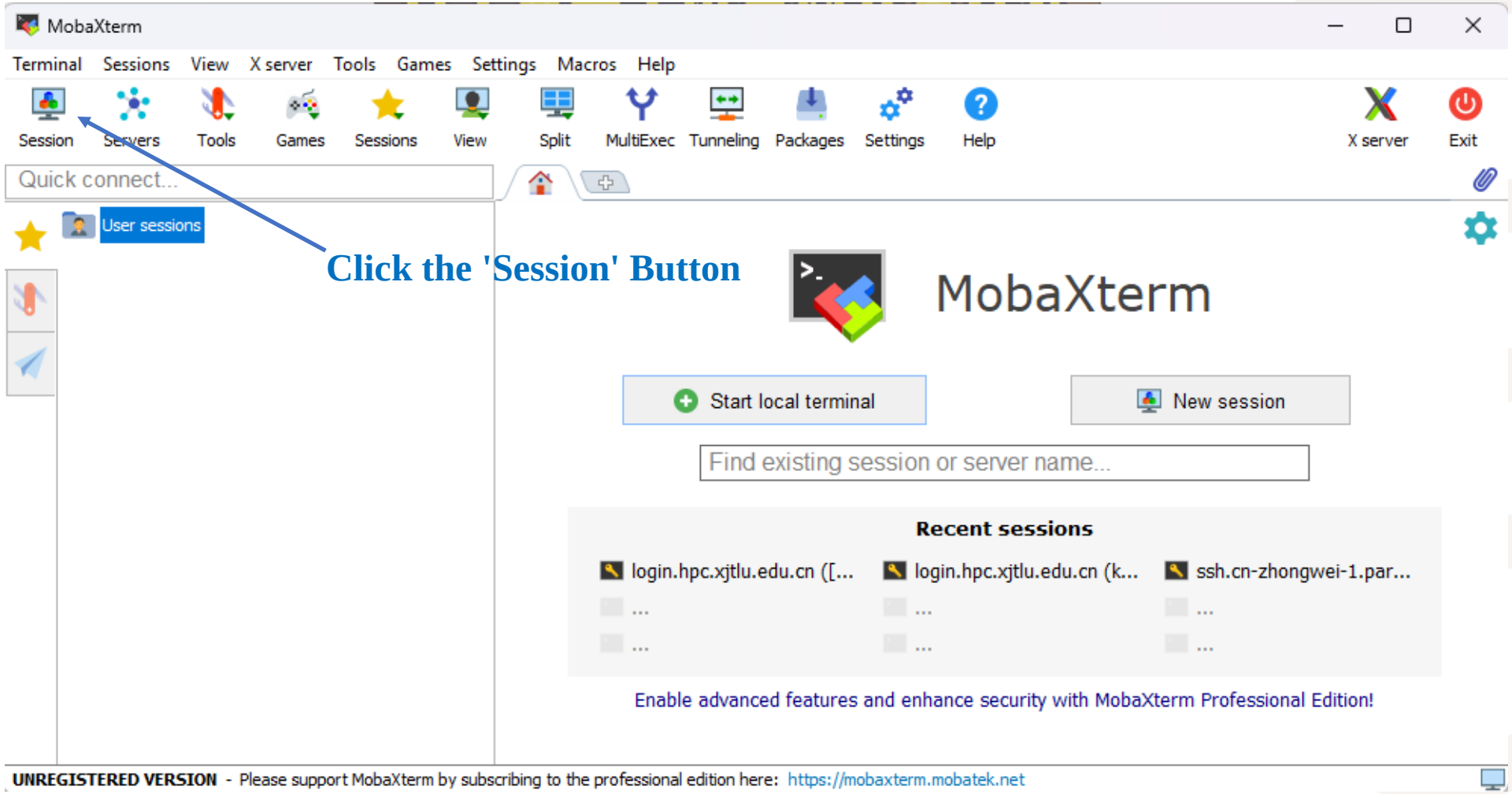
1. Install Mobaxterm software onto the computer

Official Link: https://mobaxterm.mobatek.net



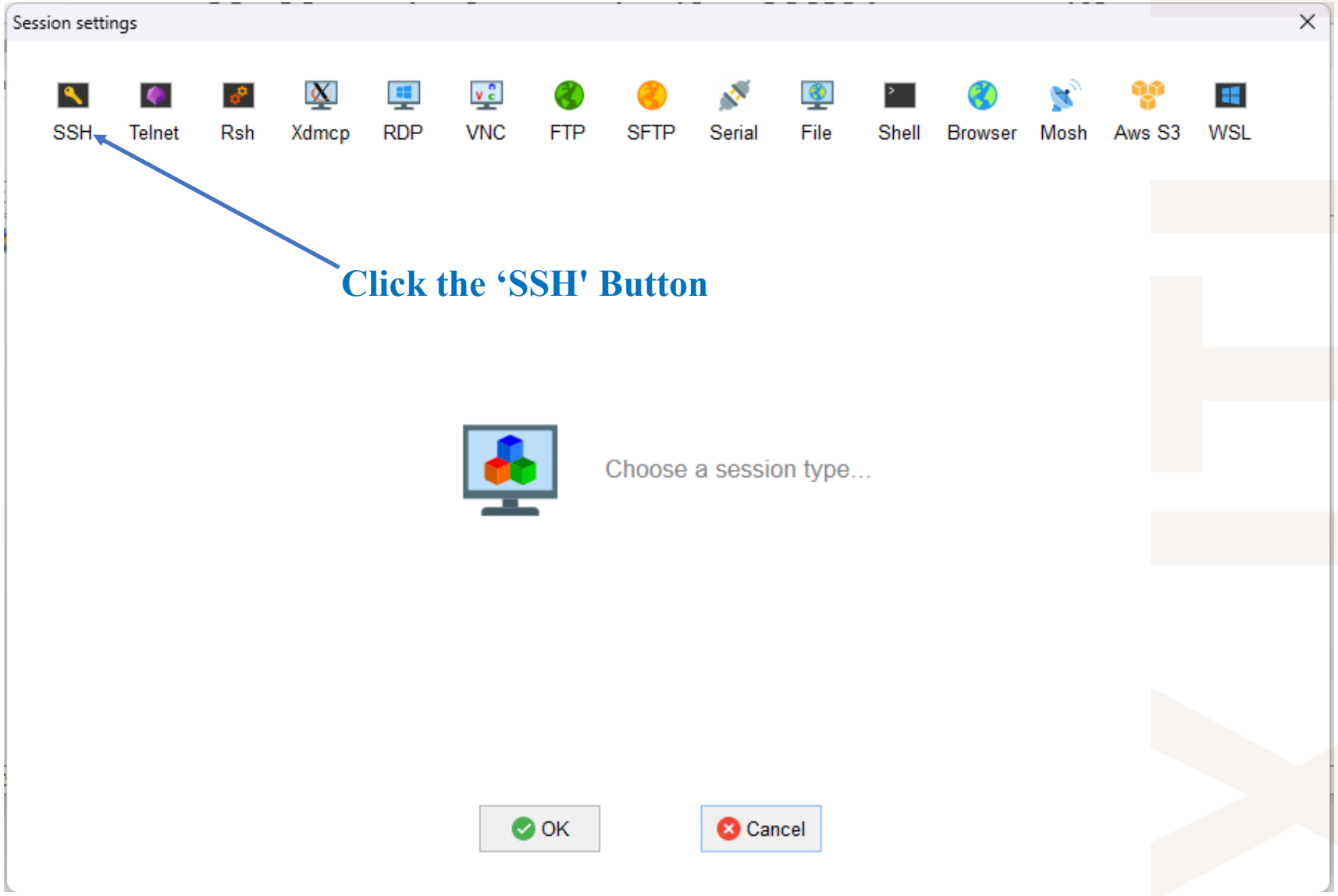
II. How to log onto the HPC account?

2. Log onto the HPC account through Mobaxterm software



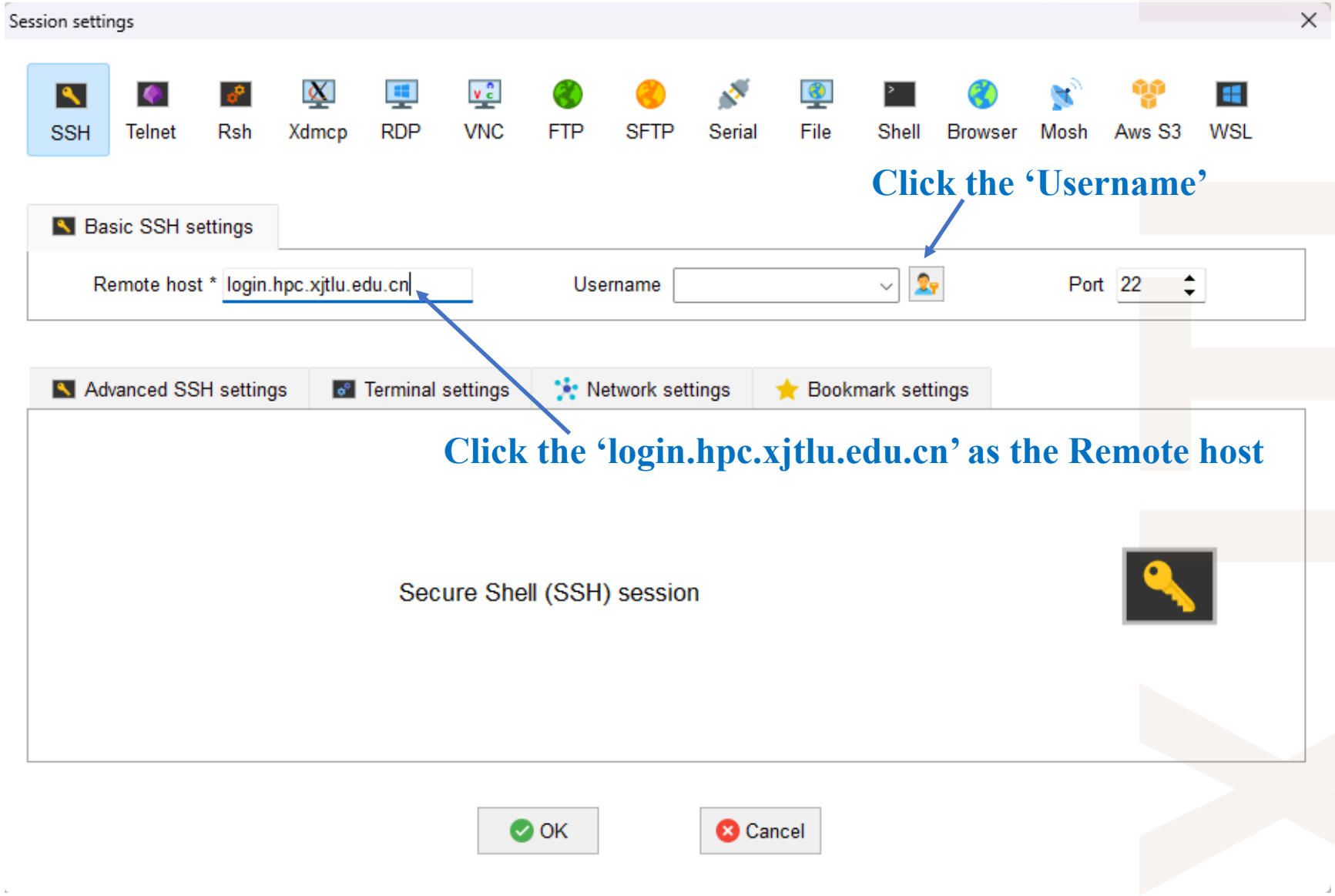
II. How to log onto the HPC account?

3. Log onto the HPC account through Mobaxterm software



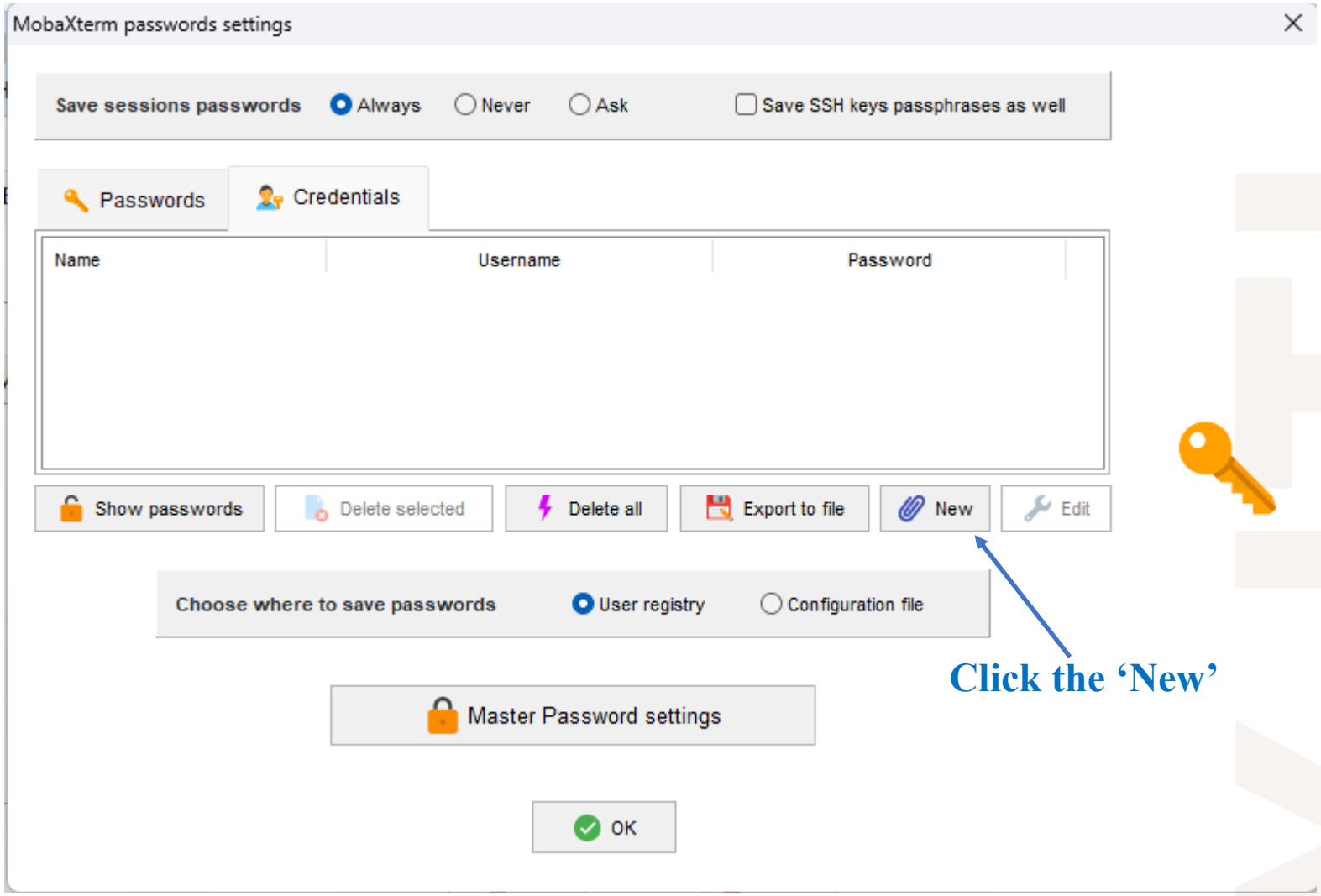
II. How to log onto the HPC account?

4. Log onto the HPC account through Mobaxterm software



II. How to log onto the HPC account?

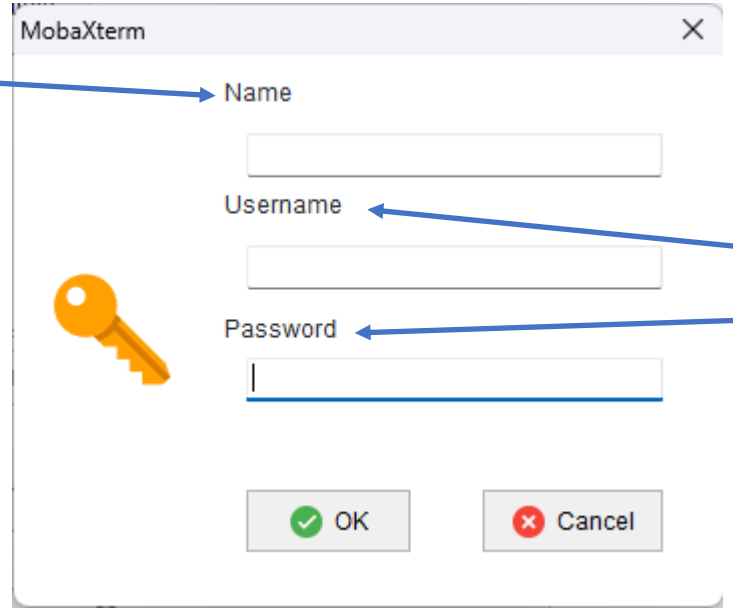
5. Log onto the HPC account through Mobaxterm software



II. How to log onto the HPC account?

6. Log onto the HPC account through Mobaxterm software

Type anything you like as the name



Type the username and password allocated to you

III. Coding for Lattice vector and *k* point in Reciprocal Space

The Reciprocal Lattice is the Fourier Transform of the Real Lattice

$$\boldsymbol{b_1 = 2\pi \frac{a_2 \times a_3}{V}}$$

$$\boldsymbol{b_2 = 2\pi \frac{a_3 \times a_1}{V}}$$

$$\boldsymbol{b_3 = 2\pi \frac{a_1 \times a_2}{V}}$$

$$\boldsymbol{V = a_1 \cdot (a_2 \times a_3) = a_2 \cdot (a_3 \times a_1) = a_3 \cdot (a_1 \times a_2)}$$

III. Coding for Lattice vector and *k* point in Reciprocal Space

Fe.vasp - Notepad

File Edit Format View Help

```
3D\Atomistic
1.0
        2.8599998951          0.0000000000          0.0000000000
        0.0000000000          2.8599998951          0.0000000000
        0.0000000000          0.0000000000          2.8599998951
   Fe
    2
Direct
     0.000000000           0.000000000           0.000000000
     0.500000000           0.500000000           0.500000000
```

$$\boldsymbol{a_1} = \begin{pmatrix} \mathbf{2.8599998951} \\ \mathbf{0} \\ \mathbf{0} \end{pmatrix}; \boldsymbol{a_2} = \begin{pmatrix} \mathbf{0} \\ \mathbf{2.8599998951} \\ \mathbf{0} \end{pmatrix}; \boldsymbol{a_3} = \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \\ \mathbf{2.8599998951} \end{pmatrix}$$

$$\boldsymbol{a_2 \times a_3 = (a_{2y} \times a_{3z} - a_{2z} \times a_{3y})\hat{x} + (a_{2z} \times a_{3x} - a_{2x} \times a_{3z})\hat{y} + (a_{2x} \times a_{3y} - a_{2y} \times a_{3x})\hat{z}}$$

$$\boldsymbol{= (2.8599998951 \times 2.8599998951 - 0 \times 0)\hat{x} + (0 \times 0 - 0 \times 2.8599998951)\hat{y} + (0 \times 0 - 2.8599998951 \times 0)\hat{z}}$$

$$\boldsymbol{= 2.8599998951^2 \hat{x}}$$

$$\boldsymbol{V = a_1 \cdot (a_2 \times a_3) = a_1 \cdot \begin{pmatrix} 2.8599998951^2 \\ 0 \\ 0 \end{pmatrix} = a_{1x} \times 2.8599998951^2 + a_{1y} \times 0 + a_{1z} \times 0}$$

$$\boldsymbol{= 2.8599998951 \times 2.8599998951^2 = 2.8599998951^3}$$

III. Coding for Lattice vector and *k* point in Reciprocal Space

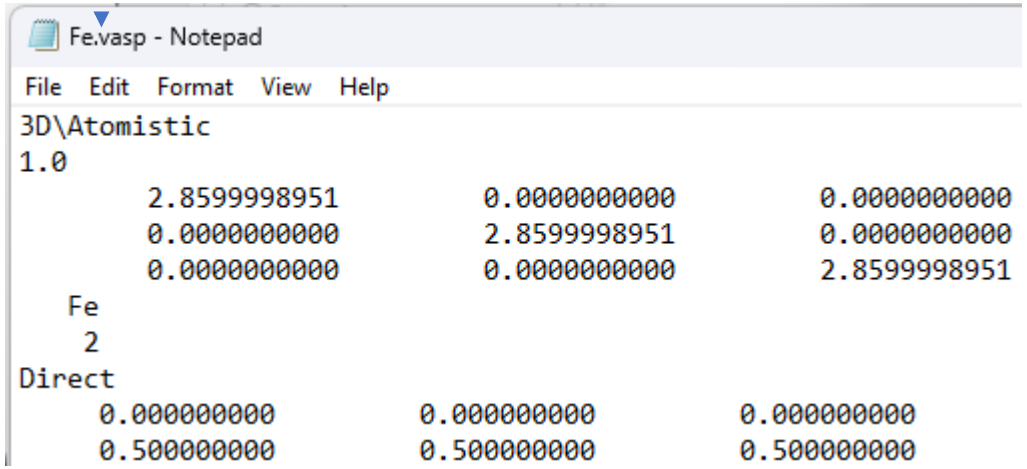
$$b_1 = 2\pi \frac{a_2 \times a_3}{V} = 2\pi \times \frac{\begin{pmatrix} 2.8599998951^2 \\ 0 \\ 0 \end{pmatrix}}{2.8599998951^3} = 2\pi \times \begin{pmatrix} 2.8599998951^{-1} \\ 0 \\ 0 \end{pmatrix}$$

$$b_2 = 2\pi \frac{a_3 \times a_1}{V} = 2\pi \times \frac{\begin{pmatrix} 0 \\ 2.8599998951^2 \\ 0 \end{pmatrix}}{2.8599998951^3} = 2\pi \times \begin{pmatrix} 0 \\ 2.8599998951^{-1} \\ 0 \end{pmatrix}$$

$$b_3 = 2\pi \frac{a_1 \times a_2}{V} = 2\pi \times \frac{\begin{pmatrix} 0 \\ 0 \\ 2.8599998951^2 \end{pmatrix}}{2.8599998951^3} = 2\pi \times \begin{pmatrix} 0 \\ 0 \\ 2.8599998951^{-1} \end{pmatrix}$$

III. Coding for Lattice vector and *k* point in Reciprocal Space

File Name



INPUT file
Contains the name of the file for reading

```
[kenansong@xpszlogin1 lecture_eight]$ cat INPUT.dat
File Name: Fe.vasp
[kenansong@xpszlogin1 lecture_eight]$
```

III. Coding for Lattice vector and *k* point in Reciprocal Space

Coding with Fortran

Structure of Fortran

- **INPUT file**
  - Contains the name of the file for reading
- **LATTICECONVERSION.f90**
  - **1. Main Part:** Read INPUT file
  - **2. Main Part:** Read Fe.vasp file
  - **3. Main Part:** Calculate and output Lattice constant in k space
    - **Call** → **Subroutine Part: KSPACE**
      Calculate the lattice constant in k space
      - **Call** → **Subroutine Part: CROSSPRODUCT**
        Calculate the cross product of two vectors
      - **Call** → **Subroutine Part: DOTPRODUCT**
        Calculate the dot product of two vectors

III. Coding for Lattice vector and *k* point in Reciprocal Space

Main Part

Read INPUT file

Read Fe.vasp file

Calculate and Lattice constant in k space

Output Lattice constant in k space

```
PROGRAM LATTICECONVERSION

IMPLICIT NONE

INTEGER, PARAMETER :: dp = SELECTED_REAL_KIND(15,14)
CHARACTER (LEN=50) :: str
INTEGER            :: i
REAL (KIND=dp)     :: latticeinrealspace(3,3)
REAL (KIND=dp)     :: latticeinkspace(3,3)

OPEN (UNIT=3, FILE='INPUT.dat', STATUS='OLD')
READ (UNIT=3, FMT='(a)') str
i = INDEX (str,':')
str = ADJUSTL(str(i+1:))
OPEN (UNIT=4, FILE=TRIM(str), STATUS='OLD')
READ (UNIT=4, FMT=*)
READ (UNIT=4, FMT=*)
DO i = 1, 3, 1
   READ (UNIT=4, FMT=*) latticeinrealspace(:,i)
END DO

CALL KSPACE(latticeinrealspace,latticeinkspace)

OPEN(UNIT=7, FILE='kspace.dat', STATUS='UNKNOWN')
DO i = 1, 3, 1
   WRITE (UNIT=7, FMT='(3X,F15.10,3X,F15.10,3X,F15.10)') latticeinkspace(:,i)
END DO

CLOSE (UNIT=3)
CLOSE (UNIT=4)

STOP
END PROGRAM LATTICECONVERSION
```

III. Coding for Lattice vector and *k* point in Reciprocal Space

Subroutine Part: CROSSPRODUCT
Calculate the cross product of two vectors

Subroutine Part: CROSSPRODUCT
Calculate the cross product of two vectors

Subroutine Part: DOTPRODUCT
Calculate the dot product of two vectors

```
SUBROUTINE KSPACE(latticeinrealspace,latticeinkspace)

IMPLICIT NONE

INTEGER, PARAMETER :: dp = SELECTED_REAL_KIND(15,14)
REAL (KIND=dp)     :: latticeinrealspace(3,3)
REAL (KIND=dp)     :: latticeinkspace(3,3)
REAL (KIND=dp)     :: volume
REAL (KIND=dp)     :: cross(3)
REAL (KIND=dp)     :: pi

pi = DACOS(-1.0d0)

CALL CROSSPRODUCT(latticeinrealspace(:,2),latticeinrealspace(:,3),cross)
CALL DOTPRODUCT(latticeinrealspace(:,1),cross,volume)

latticeinkspace(:,1) = cross / volume
CALL CROSSPRODUCT(latticeinrealspace(:,3),latticeinrealspace(:,1),cross)
latticeinkspace(:,2) = cross / volume
CALL CROSSPRODUCT(latticeinrealspace(:,1),latticeinrealspace(:,2),cross)
latticeinkspace(:,3) = cross / volume

latticeinkspace = latticeinkspace * 2.0d0 * pi

RETURN
END SUBROUTINE KSPACE
```

III. Coding for Lattice vector and *k* point in Reciprocal Space

Subroutine Part: CROSSPRODUCT
Calculate the cross product of two vectors

```
SUBROUTINE CROSSPRODUCT(vector1,vector2,result)

IMPLICIT NONE

INTEGER, PARAMETER :: dp = SELECTED_REAL_KIND(15,14)
REAL (KIND=dp)     :: vector1(3), vector2(3), result(3)

result(1) = vector1(2)*vector2(3)-vector1(3)*vector2(2)
result(2) = vector1(3)*vector2(1)-vector1(1)*vector2(3)
result(3) = vector1(1)*vector2(2)-vector1(2)*vector2(1)

RETURN
END SUBROUTINE CROSSPRODUCT
```

Subroutine Part: DOTPRODUCT
Calculate the dot product of two vectors

```
SUBROUTINE DOTPRODUCT(vector1,vector2,volume)

IMPLICIT NONE

INTEGER, PARAMETER :: dp = SELECTED_REAL_KIND(15,14)
REAL (KIND=dp)     :: vector1(3), vector2(3), volume

volume = vector1(1)*vector2(1)+vector1(2)*vector2(2)+vector1(3)*vector2(3)

RETURN
END SUBROUTINE DOTPRODUCT
```

III. Coding for Lattice vector and *k* point in Reciprocal Space

Compiling the Fortran Code

```
[kenansong@xpszlogin1 lecture_eight]$ gfortran LATTICECONVERSION.f90 -o LATTICECONVERSION
```

```
[kenansong@xpszlogin1 lecture_eight]$ ls
Fe.vasp  INPUT.dat  LATTICECONVERSION  LATTICECONVERSION.f90
```

running the Fortran Code

```
[kenansong@xpszlogin1 lecture_eight]$ ./LATTICECONVERSION
[kenansong@xpszlogin1 lecture_eight]$ ls
Fe.vasp  INPUT.dat  kspace.dat  LATTICECONVERSION  LATTICECONVERSION.f90
[kenansong@xpszlogin1 lecture_eight]$
```

III. Coding for Lattice vector and *k* point in Reciprocal Space

Content of kspace.dat

```
[kenansong@xpszlogin1 lecture_eight]$ cat kspace.dat
        2.1969180202         0.0000000000         0.0000000000
        0.0000000000         2.1969180202         0.0000000000
        0.0000000000         0.0000000000         2.1969180202
[kenansong@xpszlogin1 lecture_eight]$
```



That's all for this session

Thank you for your attention and great participation